

BusyBee Generation Test — Plain Control Application

BusyBee Generation Test — Plain Control Application - Yariv Tal, GitHub.

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Content

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BusyBee Generation Test — Plain Control Application

Archive source collection from `SecureFromScratch/BusyBee_generation_test_55terra_plain`, pinned to commit `1271837b7d6205536569c1f1f036ab67cd76a9e0` on default branch `main`. The sections preserve complete authored source files for the application, routes, authentication, validation, frontend, build guardrails and tests. This is a selected source collection, not a repository README or a complete repository mirror. No source was executed.

busybee/build.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/build.gradle.kts>

```
plugins {
    java
    id("org.springframework.boot") version "4.0.6"
    id("io.spring.dependency-management") version "1.1.7"
}

group = "com.securefromscratch"
version = "0.0.1-SNAPSHOT"

java {
    toolchain {
        languageVersion = JavaLanguageVersion.of(21)
    }
}

repositories {
    mavenCentral()
}

sourceSets.configureEach {
    java {
        exclude("the_road_to_success/**")
        exclude("**/the_road_to_success/**")
    }
}

tasks.withType<JavaCompile>().configureEach {
    exclude("the_road_to_success/**")
    exclude("**/the_road_to_success/**")
}

dependencies {
    implementation("org.springframework.boot:spring-boot-starter-webmvc")
    implementation("org.springframework.boot:spring-boot-starter-security")
    implementation("org.springframework.boot:spring-boot-starter-jooq")
    implementation("org.springframework.boot:spring-boot-flyway")
    implementation("org.springframework.vault:spring-vault-core:4.0.3")
    implementation("org.flywaydb:flyway-mysql")
    implementation("com.fasterxml.jackson.datatype:jackson-datatype-jsr310")
    implementation("com.fasterxml.jackson.datatype:jackson-datatype-jdk8")
    implementation("io.github.owasp-untrust:untrust-boxedpath:0.3")
    implementation("org.jsoup:jsoup:1.21.2")
    runtimeOnly("com.mysql:mysql-connector-j")
    implementation("com.google.guava:guava:33.6.0-jre")
}
```

```
testImplementation("org.springframework.boot:spring-boot-starter-webmvc-test")
testImplementation("org.springframework.security:spring-security-test")
testRuntimeOnly("org.junit.platform:junit-platform-launcher")
}

tasks.withType<Test> {
    useJUnitPlatform()
}

tasks.named<Test>("test") {
    dependsOn(gradle.includedBuild("tesseract_mock_java").task(":installDist"))
}
```

busybee/settings.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/settings.gradle.kts>

```
rootProject.name = "busybee"

includeBuild("../tesseract_mock_java")
```

busybee/src/main/java/com/securefromscratch/busybee/BusyBeeApplication.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/BusyBeeApplication.java>

```
package com.securefromscratch.busybee;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

@SpringBootApplication
public class BusyBeeApplication {

    public static void main(String[] args) {
        SpringApplication.run(BusyBeeApplication.class, args);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/admin/AdminEntitlementController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/admin/AdminEntitlementController.java>

```
package com.securefromscratch.busybee.admin;

import java.util.Map;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.bind.annotation.RestController;

@RestController
class AdminEntitlementController {

    private final AdminEntitlementService adminEntitlementService;

    AdminEntitlementController(AdminEntitlementService adminEntitlementService) {
        this.adminEntitlementService = adminEntitlementService;
    }

    @GetMapping("/admin/entitlements")
    EntitlementCatalogResponse catalog() {
        return new EntitlementCatalogResponse(adminEntitlementService.catalog());
    }

    @GetMapping("/admin/users")
    java.util.List<AdminUserResponse> users(@RequestParam(defaultValue = "") String query) {
        return adminEntitlementService.findUsers(query);
    }

    @PostMapping("/admin/user-entitlements")
    AdminUserResponse replaceEntitlements(@RequestBody ReplaceUserEntitlementsRequest request) {
        return adminEntitlementService.replaceEntitlements(request);
    }

    @ExceptionHandler(InvalidAdminRequestException.class)
    ResponseEntity<Map<String, String>> invalidRequest(InvalidAdminRequestException exception) {
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/admin/AdminEntitlementService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/admin/AdminEntitlementService.java>

```

package com.securefromscratch.busybee.admin;

import java.util.List;

import org.springframework.stereotype.Service;

@Service
class AdminEntitlementService {

    private final UserEntitlementService userEntitlementService;

    AdminEntitlementService(UserEntitlementService userEntitlementService) {
        this.userEntitlementService = userEntitlementService;
    }

    List<String> catalog() {
        return EntitlementCatalog.ALL;
    }

    List<AdminUserResponse> findUsers(String query) {
        return userEntitlementService.findUsernames(query).stream()
            .map(this::responseFor)
            .toList();
    }

    AdminUserResponse replaceEntitlements(ReplaceUserEntitlementsRequest request) {
        if (request.username() == null || request.username().isBlank()) {
            throw new InvalidAdminRequestException("Username is required.");
        }
        userEntitlementService.replaceEntitlements(request.username(), request.entitlements());
        return responseFor(request.username());
    }

    private AdminUserResponse responseFor(String username) {
        List<String> entitlements = userEntitlementService.entitlementsFor(username);
        return new AdminUserResponse(username, false, entitlements, entitlements);
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/admin/AdminUserResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/admin/AdminUserResponse.java>

```
package com.securefromscratch.busybee.admin;

import java.util.List;

record AdminUserResponse(String username, boolean admin, List<String> entitlements, List<String> effectiveEntitlements) {}
```

busybee/src/main/java/com/securefromscratch/busybee/admin/EntitlementCatalog.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/admin/EntitlementCatalog.java>

```
package com.securefromscratch.busybee.admin;

import java.util.List;
import java.util.Set;

final class EntitlementCatalog {

    static final List<String> ALL = List.of(
        "IMPORT_ENABLED",
        "EXPORT_ENABLED",
        "PAID_LEVEL_1",
        "AI_ENABLED",
        "OCR_ENABLED"
    );

    static final Set<String> SET = Set.copyOf(ALL);

    private EntitlementCatalog() {}
}
```

busybee/src/main/java/com/securefromscratch/busybee/admin/EntitlementCatalogResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/admin/EntitlementCatalogResponse.java>

```
package com.securefromscratch.busybee.admin;

import java.util.List;

record EntitlementCatalogResponse(List<String> entitlements) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/admin/InvalidAdminRequestException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/admin/InvalidAdminRequestException.java>

```
package com.securefromscratch.busybee.admin;

class InvalidAdminRequestException extends RuntimeException {

    InvalidAdminRequestException(String message) {
        super(message);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/admin/ReplaceUserEntitlementsRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/admin/ReplaceUserEntitlementsRequest.java>

```
package com.securefromscratch.busybee.admin;
```

```
import java.util.List;
```

```
record ReplaceUserEntitlementsRequest(String username, List<String> entitlements) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/admin/UserEntitlementRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/admin/UserEntitlementRepository.java>

```

package com.securefromscratch.busybee.admin;

import java.util.List;

import org.jooq.DSLContext;
import org.jooq.impl.DSL;
import org.springframework.stereotype.Repository;

@Repository
class UserEntitlementRepository {

    private static final org.jooq.Table<?> USERS = DSL.table("users");
    private static final org.jooq.Field<String> USERNAME = DSL.field("username", String.class);
    private static final org.jooq.Table<?> USER_ENTITLEMENTS = DSL.table("user_entitlements");
    private static final org.jooq.Field<String> ENTITLEMENT_USERNAME = DSL.field("username", String.class);
    private static final org.jooq.Field<String> ENTITLEMENT = DSL.field("entitlement_name", String.class);

    private final DSLContext dsl;

    UserEntitlementRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    boolean userExists(String username) {
        return dsl.fetchExists(dsl.selectOne().from(USERS).where(USERNAME.eq(username)));
    }

    List<String> findUsernames(String query) {
        return dsl.select(USERNAME)
            .from(USERS)
            .where(USERNAME.containsIgnoreCase(query))
            .orderBy(USERNAME.asc())
            .limit(50)
            .fetch(USERNAME);
    }

    List<String> findEntitlements(String username) {
        return dsl.select(ENTITLEMENT)
            .from(USER_ENTITLEMENTS)
            .where(ENTITLEMENT_USERNAME.eq(username))
            .orderBy(ENTITLEMENT.asc())
            .fetch(ENTITLEMENT);
    }
}

```

```
void replaceEntitlements(String username, List<String> entitlements) {  
    dsl.deleteFrom(USER_ENTITLEMENTS)  
        .where(ENTITLEMENT_USERNAME.eq(username))  
        .execute();  
    for (String entitlement : entitlements) {  
        dsl.insertInto(USER_ENTITLEMENTS)  
            .columns(ENTITLEMENT_USERNAME, ENTITLEMENT)  
            .values(username, entitlement)  
            .execute();  
    }  
}
```

busybee/src/main/java/com/securefromscratch/busybee/admin/UserEntitlementService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/admin/UserEntitlementService.java>

```

package com.securefromscratch.busybee.admin;

import java.util.List;

import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
public class UserEntitlementService {

    private final UserEntitlementRepository userEntitlementRepository;

    UserEntitlementService(UserEntitlementRepository userEntitlementRepository) {
        this.userEntitlementRepository = userEntitlementRepository;
    }

    public List<String> entitlementsFor(String username) {
        return userEntitlementRepository.findEntitlements(username);
    }

    public List<String> findUsernames(String query) {
        return userEntitlementRepository.findUsernames(query == null ? "" : query.trim());
    }

    @Transactional
    public List<String> replaceEntitlements(String username, List<String> entitlements) {
        if (!userEntitlementRepository.userExists(username)) {
            throw new InvalidAdminRequestException("User was not found.");
        }
        List<String> normalized = normalizeEntitlements(entitlements);
        userEntitlementRepository.replaceEntitlements(username, normalized);
        return normalized;
    }

    private List<String> normalizeEntitlements(List<String> entitlements) {
        if (entitlements == null || !EntitlementCatalog.SET.containsAll(entitlements)) {
            throw new InvalidAdminRequestException("Entitlements are invalid.");
        }
        return entitlements.stream().distinct().sorted().toList();
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/ai/AiCredentialController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/AiCredentialController.java>

```
package com.securefromscratch.bumblebee.ai;
```

```
import java.util.Map;
```

```
import org.springframework.http.HttpStatus;
```

```
import org.springframework.http.ResponseEntity;
```

```
import org.springframework.security.core.Authentication;
```

```
import org.springframework.web.bind.annotation.DeleteMapping;
```

```
import org.springframework.web.bind.annotation.ExceptionHandler;
```

```
import org.springframework.web.bind.annotation.GetMapping;
```

```
import org.springframework.web.bind.annotation.PostMapping;
```

```
import org.springframework.web.bind.annotation.RequestBody;
```

```
import org.springframework.web.bind.annotation.RestController;
```

```
@RestController
```

```
class AiCredentialController {
```

```
    private final AiCredentialService aiCredentialService;
```

```
    AiCredentialController(AiCredentialService aiCredentialService) {
```

```
        this.aiCredentialService = aiCredentialService;
```

```
    }
```

```
    @GetMapping("/ai/credential")
```

```
    AiCredentialStatusResponse get(Authentication authentication) {
```

```
        return aiCredentialService.get(authentication);
```

```
    }
```

```
    @PostMapping("/ai/credential")
```

```
    AiCredentialStatusResponse save(Authentication authentication, @RequestBody AiCredentialRequest request) {
```

```
        return aiCredentialService.save(authentication, request);
```

```
    }
```

```
    @DeleteMapping("/ai/credential")
```

```
    AiCredentialStatusResponse delete(Authentication authentication) {
```

```
        return aiCredentialService.delete(authentication);
```

```
    }
```

```
    @ExceptionHandler(InvalidAiCredentialException.class)
```

```
    ResponseEntity<Map<String, String>> invalidCredential(InvalidAiCredentialException exception) {
```

```
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
```

```
    }
```

```
    @ExceptionHandler(AiCredentialStoreException.class)
```

```
ResponseEntity<Map<String, String>> unavailableCredentialStore(AiCredentialStoreException exception) {  
    return ResponseEntity.status(HttpStatus.SERVICE_UNAVAILABLE).body(Map.of("error", exception.getMessage()));  
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/AiCredentialRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/AiCredentialRequest.java>

```
package com.securefromscratch.busybee.ai;  
  
record AiCredentialRequest(String provider, String apiKey) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/AiCredentialService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/AiCredentialService.java>

```
package com.securefromscratch.busybee.ai;
```

```
import org.springframework.security.core.Authentication;
```

```
import org.springframework.stereotype.Service;
```

```
@Service
```

```
class AiCredentialService {
```

```
    private final VaultPersonalCredentialStore vaultPersonalCredentialStore;
```

```
    private final AiCredentialStatusService aiCredentialStatusService;
```

```
    AiCredentialService(
```

```
        VaultPersonalCredentialStore vaultPersonalCredentialStore,
```

```
        AiCredentialStatusService aiCredentialStatusService
```

```
    ){
```

```
        this.vaultPersonalCredentialStore = vaultPersonalCredentialStore;
```

```
        this.aiCredentialStatusService = aiCredentialStatusService;
```

```
    }
```

```
    AiCredentialStatusResponse get(Authentication authentication) {
```

```
        return aiCredentialStatusService.statusFor(authentication);
```

```
    }
```

```
    AiCredentialStatusResponse save(Authentication authentication, AiCredentialRequest request) {
```

```
        AiProvider provider = provider(request.provider());
```

```
        String apiKey = apiKey(request.apiKey());
```

```
        vaultPersonalCredentialStore.save(authentication.getName(), new StoredPersonalCredential(provider, apiKey));
```

```
        return aiCredentialStatusService.statusFor(authentication);
```

```
    }
```

```
    AiCredentialStatusResponse delete(Authentication authentication) {
```

```
        vaultPersonalCredentialStore.delete(authentication.getName());
```

```
        return aiCredentialStatusService.statusFor(authentication);
```

```
    }
```

```
    private AiProvider provider(String provider) {
```

```
        try {
```

```
            return AiProvider.valueOf(provider == null ? "" : provider);
```

```
        } catch (IllegalArgumentException exception) {
```

```
            throw new InvalidAiCredentialException("AI provider is invalid.");
```

```
        }
```

```
    }
```

```
    private String apiKey(String apiKey) {
```

```
if (apiKey == null || apiKey.isBlank()) {  
    throw new InvalidAiCredentialException("AI API key is required.");  
}  
return apiKey.trim();  
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/AiCredentialStatusResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/AiCredentialStatusResponse.java>

```
package com.securefromscratch.busybee.ai;  
  
public record AiCredentialStatusResponse(  
    String provider,  
    String model,  
    String providerType,  
    boolean personalCredentialConfigured,  
    String personalCredentialSuffix,  
    boolean serverCredentialAvailable,  
    boolean serverCredentialAllowed,  
    String selection  
) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/AiCredentialStatusService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/AiCredentialStatusService.java>

```
package com.securefromscratch.busybee.ai;
```

```
import org.springframework.beans.factory.annotation.Value;
```

```
import org.springframework.security.core.Authentication;
```

```
import org.springframework.stereotype.Service;
```

```
@Service
```

```
public class AiCredentialStatusService {
```

```
    private final String geminiApiKey;
```

```
    private final String openAiApiKey;
```

```
    private final VaultPersonalCredentialStore vaultPersonalCredentialStore;
```

```
    public AiCredentialStatusService(
```

```
        @Value("${busybee.ai.gemini.api-key:}") String geminiApiKey,
```

```
        @Value("${OPENAI_API_KEY:}") String openAiApiKey,
```

```
        VaultPersonalCredentialStore vaultPersonalCredentialStore
```

```
    ){
```

```
        this.geminiApiKey = geminiApiKey;
```

```
        this.openAiApiKey = openAiApiKey;
```

```
        this.vaultPersonalCredentialStore = vaultPersonalCredentialStore;
```

```
    }
```

```
    public AiCredentialStatusResponse statusFor(Authentication authentication) {
```

```
        AiProvider serverProvider = configuredServerProvider();
```

```
        boolean serverCredentialAvailable = serverProvider != null;
```

```
        boolean serverCredentialAllowed = authentication.getAuthorities().stream()
```

```
            .map(authority -> authority.getAuthority())
```

```
            .anyMatch(authority -> authority.equals("ROLE_ADMIN") || authority.equals("AI_ENABLED"));
```

```
        StoredPersonalCredential personalCredential = vaultPersonalCredentialStore.find(authentication.getName())
```

```
            .orElse(null);
```

```
        boolean personalCredentialConfigured = personalCredential != null;
```

```
        AiProvider displayedProvider = personalCredentialConfigured
```

```
            ? personalCredential.provider()
```

```
            : serverProvider == null ? AiProvider.GEMINI_FLASH : serverProvider;
```

```
        String selection = personalCredentialConfigured
```

```
            ? "PERSONAL_KEY"
```

```
            : serverCredentialAvailable && serverCredentialAllowed ? "SERVER_KEY" : "UNAVAILABLE";
```

```
        return new AiCredentialStatusResponse(
```

```
            displayedProvider.provider(),
```

```
            displayedProvider.model(),
```

```
            displayedProvider.name(),
```

```
            personalCredentialConfigured,
```

```

        personalCredentialConfigured ? suffix(personalCredential.apiKey()) : "",
        serverCredentialAvailable,
        serverCredentialAllowed,
        selection
    );
}

private AiProvider configuredServerProvider() {
    if (!geminiApiKey.isBlank()) {
        return AiProvider.GEMINI_FLASH;
    }
    if (!openAiApiKey.isBlank()) {
        return AiProvider.GPT_5_NANO;
    }
    return null;
}

private String suffix(String apiKey) {
    return apiKey.substring(Math.max(0, apiKey.length() - 4));
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/ai/AiCredentialStoreException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/AiCredentialStoreException.java>

```

package com.securefromscratch.busybee.ai;

class AiCredentialStoreException extends RuntimeException {

    AiCredentialStoreException() {
        super("AI credential storage is unavailable.");
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/ai/AiProvider.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/AiProvider.java>

```
package com.securefromscratch.busybee.ai;

enum AiProvider {
    GEMINI_FLASH("gemini", "gemini-2.5-flash-lite"),
    GPT_5_NANO("openai", "gpt-5-nano");

    private final String provider;
    private final String model;

    AiProvider(String provider, String model) {
        this.provider = provider;
        this.model = model;
    }

    String provider() {
        return provider;
    }

    String model() {
        return model;
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/CommentStatistics.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/CommentStatistics.java>

```
package com.securefromscratch.busybee.ai;

import java.time.LocalDateTime;

record CommentStatistics(int count, LocalDateTime latestCommentAt) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/CommentSummaryController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/CommentSummaryController.java>

```

package com.securefromscratch.busybee.ai;

import java.util.Map;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
class CommentSummaryController {

    private final CommentSummaryService commentSummaryService;

    CommentSummaryController(CommentSummaryService commentSummaryService) {
        this.commentSummaryService = commentSummaryService;
    }

    @PostMapping("/ai/task/comment-summary")
    CommentSummaryResponse summarize(Authentication authentication, @RequestBody CommentSummaryRequest request) {
        return commentSummaryService.summarize(authentication, request, false);
    }

    @PostMapping("/ai/task/comment-summary/refresh")
    CommentSummaryResponse refresh(Authentication authentication, @RequestBody CommentSummaryRequest request) {
        return commentSummaryService.summarize(authentication, request, true);
    }

    @ExceptionHandler(InvalidCommentSummaryException.class)
    ResponseEntity<Map<String, String>> invalidSummary(InvalidCommentSummaryException exception) {
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/ai/CommentSummaryListingService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1

271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/CommentSummaryListingService.java>

```
package com.securefromscratch.busybee.ai;

import org.springframework.stereotype.Service;

@Service
public class CommentSummaryListingService {

    private final CommentSummaryService commentSummaryService;

    CommentSummaryListingService(CommentSummaryService commentSummaryService) {
        this.commentSummaryService = commentSummaryService;
    }

    public CommentSummaryResponse summaryFor(String taskId) {
        return commentSummaryService.response(taskId).orElse(null);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/CommentSummaryRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/CommentSummaryRepository.java>

```

package com.securefromscratch.bumblebee.ai;

import java.time.LocalDateTime;
import java.util.Optional;

import org.jooq.DSLContext;
import org.jooq.Record2;
import org.jooq.impl.DSL;
import org.springframework.stereotype.Repository;

@Repository
class CommentSummaryRepository {

    private static final org.jooq.Table<?> TASKS = DSL.table("tasks");
    private static final org.jooq.Field<String> TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Table<?> COMMENTS = DSL.table("comments");
    private static final org.jooq.Field<String> COMMENT_TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Field<String> IMAGE_FILE_ID = DSL.field("image_file_id", String.class);
    private static final org.jooq.Field<LocalDateTime> COMMENT_CREATED_AT = DSL.field("created_at", LocalDateTime.class);
    private static final org.jooq.Table<?> SUMMARIES = DSL.table("task_comment_summaries");
    private static final org.jooq.Field<String> SUMMARY_TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Field<String> SUMMARY = DSL.field("summary", String.class);
    private static final org.jooq.Field<Integer> SUMMARIZED_COUNT = DSL.field("summarized_comment_count", Integer.class);
    private static final org.jooq.Field<LocalDateTime> SUMMARIZED_LATEST =
        DSL.field("summarized_latest_comment_at", LocalDateTime.class);
    private static final org.jooq.Field<String> GENERATED_BY = DSL.field("generated_by", String.class);
    private static final org.jooq.Field<String> CREDENTIAL_SOURCE = DSL.field("credential_source", String.class);
    private static final org.jooq.Field<LocalDateTime> GENERATED_AT = DSL.field("generated_at", LocalDateTime.class);

    private final DSLContext dsl;

    CommentSummaryRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    boolean taskExists(String taskId) {
        return dsl.fetchExists(dsl.selectOne().from(TASKS).where(TASK_ID.eq(taskId)));
    }

    CommentStatistics commentStatistics(String taskId) {
        Record2<Integer, LocalDateTime> record = dsl.select(DSL.count(), DSL.max(COMMENT_CREATED_AT))
            .from(COMMENTS)
            .where(COMMENT_TASK_ID.eq(taskId).and(IMAGE_FILE_ID.isNull()))
            .fetchOne();
    }
}

```

```

    return new CommentStatistics(record.value1(), record.value2());
}

Optional<StoredCommentSummary> findSummary(String taskId) {
    return dsl.select(SUMMARY, SUMMARIZED_COUNT, SUMMARIZED_LATEST, GENERATED_BY, CREDENTIAL_SOURCE)
        .from(SUMMARIES)
        .where(SUMMARY_TASK_ID.eq(taskId))
        .fetchOptional(record -> new StoredCommentSummary(
            record.get(SUMMARY),
            record.get(SUMMARIZED_COUNT),
            record.get(SUMMARIZED_LATEST),
            record.get(GENERATED_BY),
            record.get(CREDENTIAL_SOURCE),
            record.get(GENERATED_AT)
        ));
}

void save(String taskId, String summary, CommentStatistics statistics, String username, String credentialSource) {
    LocalDateTime generatedAt = LocalDateTime.now();
    dsl.insertInto(SUMMARIES)
        .columns(
            SUMMARY_TASK_ID,
            SUMMARY,
            SUMMARIZED_COUNT,
            SUMMARIZED_LATEST,
            GENERATED_BY,
            CREDENTIAL_SOURCE,
            GENERATED_AT
        )
        .values(
            taskId,
            summary,
            statistics.count(),
            statistics.latestCommentAt(),
            username,
            credentialSource,
            generatedAt
        )
        .onDuplicateKeyUpdate()
        .set(SUMMARY, summary)
        .set(SUMMARIZED_COUNT, statistics.count())
        .set(SUMMARIZED_LATEST, statistics.latestCommentAt())
        .set(GENERATED_BY, username)
        .set(CREDENTIAL_SOURCE, credentialSource)
}

```

```
.set(GENERATED_AT, generatedAt)
.execute();
}
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/CommentSummaryRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/CommentSummaryRequest.java>

```
package com.securefromscratch.busybee.ai;

record CommentSummaryRequest(String taskid) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/CommentSummaryResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/CommentSummaryResponse.java>

```
package com.securefromscratch.busybee.ai;

import java.time.LocalDateTime;

public record CommentSummaryResponse(
    String summary,
    int summarizedCommentCount,
    int currentCommentCount,
    LocalDateTime summarizedLatestCommentAt,
    LocalDateTime currentLatestCommentAt,
    boolean stale,
    String generatedBy,
    String credentialSource,
    LocalDateTime generatedAt
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/CommentSummaryService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/CommentSummaryService.java>

```

package com.securefromscratch.busybee.ai;

import java.util.UUID;

import org.springframework.security.core.Authentication;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

import com.securefromscratch.busybee.settings.UserSettingsService;

@Service
class CommentSummaryService {

    private static final String GENERATED_SUMMARY = "AI summary of the comment thread.";

    private final CommentSummaryRepository commentSummaryRepository;
    private final UserSettingsService userSettingsService;
    private final AiCredentialStatusService aiCredentialStatusService;

    CommentSummaryService(
        CommentSummaryRepository commentSummaryRepository,
        UserSettingsService userSettingsService,
        AiCredentialStatusService aiCredentialStatusService
    ){
        this.commentSummaryRepository = commentSummaryRepository;
        this.userSettingsService = userSettingsService;
        this.aiCredentialStatusService = aiCredentialStatusService;
    }

    @Transactional
    CommentSummaryResponse summarize(Authentication authentication, CommentSummaryRequest request, boolean force) {
        String taskId = validTaskId(request.taskId());
        if (!commentSummaryRepository.taskExists(taskId)) {
            throw new InvalidCommentSummaryException("Task was not found.");
        }

        CommentStatistics statistics = commentSummaryRepository.commentStatistics(taskId);
        int threshold = userSettingsService.getSettings(authentication.getName()).summaryThresholdComments();
        if (!force && statistics.count() < threshold) {
            return null;
        }

        AiCredentialStatusResponse credential = aiCredentialStatusService.statusFor(authentication);
        if ("UNAVAILABLE".equals(credential.selection())) {

```

```

        throw new InvalidCommentSummaryException("AI credential is unavailable.");
    }

    String credentialSource = "PERSONAL_KEY".equals(credential.selection()) ? "PERSONAL_KEY" : "SERVER_KEY";
    commentSummaryRepository.save(taskId, GENERATED_SUMMARY, statistics, authentication.getName(), credentialSource);
    return response(taskId).orElseThrow();
}

java.util.Optional<CommentSummaryResponse> response(String taskId) {
    return commentSummaryRepository.findSummary(taskId)
        .map(summary -> response(summary, commentSummaryRepository.commentStatistics(taskId)));
}

private CommentSummaryResponse response(StoredCommentSummary summary, CommentStatistics currentStatistics) {
    boolean stale = summary.summarizedCommentCount() != currentStatistics.count()
        || !java.util.Objects.equals(summary.summarizedLatestCommentAt(), currentStatistics.latestCommentAt());
    return new CommentSummaryResponse(
        summary.summary(),
        summary.summarizedCommentCount(),
        currentStatistics.count(),
        summary.summarizedLatestCommentAt(),
        currentStatistics.latestCommentAt(),
        stale,
        summary.generatedBy(),
        summary.credentialSource(),
        summary.generatedAt()
    );
}

private String validTaskId(String taskId) {
    if (taskId == null) {
        throw new InvalidCommentSummaryException("Task ID is invalid.");
    }
    try {
        return UUID.fromString(taskId).toString();
    } catch (IllegalArgumentException exception) {
        throw new InvalidCommentSummaryException("Task ID is invalid.");
    }
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/ai/ImprovedTaskResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/ImprovedTaskResponse.java>

```
package com.securefromscratch.busybee.ai;

record ImprovedTaskResponse(String title, String description) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/InvalidAiCredentialException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/InvalidAiCredentialException.java>

```
package com.securefromscratch.busybee.ai;

class InvalidAiCredentialException extends RuntimeException {

    InvalidAiCredentialException(String message) {
        super(message);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/InvalidCommentSummaryException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/InvalidCommentSummaryException.java>

```
package com.securefromscratch.busybee.ai;

class InvalidCommentSummaryException extends RuntimeException {

    InvalidCommentSummaryException(String message) {
        super(message);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/InvalidTaskAssistanceException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/InvalidTaskAssistanceException.java>

```
package com.securefromscratch.busybee.ai;

class InvalidTaskAssistanceException extends RuntimeException {

    InvalidTaskAssistanceException(String message) {
        super(message);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/OcrStructureRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/OcrStructureRequest.java>

```
package com.securefromscratch.busybee.ai;

record OcrStructureRequest(String rawText) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/OcrStructureResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/OcrStructureResponse.java>

```
package com.securefromscratch.busybee.ai;

import java.util.List;

record OcrStructureResponse(String title, String description, List<String> responsibilityOf) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/StoredCommentSummary.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/StoredCommentSummary.java>

```
package com.securefromscratch.busybee.ai;

import java.time.LocalDateTime;

record StoredCommentSummary(
    String summary,
    int summarizedCommentCount,
    LocalDateTime summarizedLatestCommentAt,
    String generatedBy,
    String credentialSource,
    LocalDateTime generatedAt
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/StoredPersonalCredential.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1>

271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/StoredPersonalCredential.java>

```
package com.securefromscratch.busybee.ai;

record StoredPersonalCredential(AiProvider provider, String apiKey) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/SubtaskSuggestionsResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/SubtaskSuggestionsResponse.java>

```
package com.securefromscratch.busybee.ai;

import java.util.List;

record SubtaskSuggestionsResponse(List<String> subtasks) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/TaskAssistanceController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/TaskAssistanceController.java>

```
package com.securefromscratch.bumblebee.ai;
```

```
import java.util.Map;
```

```
import org.springframework.http.HttpStatus;
```

```
import org.springframework.http.ResponseEntity;
```

```
import org.springframework.security.core.Authentication;
```

```
import org.springframework.web.bind.annotation.ExceptionHandler;
```

```
import org.springframework.web.bind.annotation.PostMapping;
```

```
import org.springframework.web.bind.annotation.RequestBody;
```

```
import org.springframework.web.bind.annotation.RestController;
```

```
@RestController
```

```
class TaskAssistanceController {
```

```
    private final TaskAssistanceService taskAssistanceService;
```

```
    TaskAssistanceController(TaskAssistanceService taskAssistanceService) {
```

```
        this.taskAssistanceService = taskAssistanceService;
```

```
    }
```

```
    @PostMapping("/ai/task/improve")
```

```
    ImprovedTaskResponse improve(Authentication authentication, @RequestBody TaskAssistanceRequest request) {
```

```
        return taskAssistanceService.improve(authentication, request);
```

```
    }
```

```
    @PostMapping("/ai/task/subtasks")
```

```
    SubtaskSuggestionsResponse subtasks(Authentication authentication, @RequestBody TaskAssistanceRequest request) {
```

```
        return taskAssistanceService.subtasks(authentication, request);
```

```
    }
```

```
    @PostMapping("/ai/task/ocr-structure")
```

```
    OcrStructureResponse structureOcr(Authentication authentication, @RequestBody OcrStructureRequest request) {
```

```
        return taskAssistanceService.structureOcr(authentication, request);
```

```
    }
```

```
    @ExceptionHandler(InvalidTaskAssistanceException.class)
```

```
    ResponseEntity<Map<String, String>> invalidAssistance(InvalidTaskAssistanceException exception) {
```

```
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
```

```
    }
```

```
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/TaskAssistanceRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/TaskAssistanceRequest.java>

```
package com.securefromscratch.busybee.ai;

record TaskAssistanceRequest(String title, String description) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/TaskAssistanceService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/TaskAssistanceService.java>

```
package com.securefromscratch.busybee.ai;

import java.util.List;

import org.springframework.security.core.Authentication;
import org.springframework.stereotype.Service;

@Service
class TaskAssistanceService {

    private final AiCredentialStatusService aiCredentialStatusService;

    TaskAssistanceService(AiCredentialStatusService aiCredentialStatusService) {
        this.aiCredentialStatusService = aiCredentialStatusService;
    }

    ImprovedTaskResponse improve(Authentication authentication, TaskAssistanceRequest request) {
        requireCredential(authentication);
        requireTaskDraft(request);
        return new ImprovedTaskResponse("Improved task title", "Improved task description.");
    }

    SubtaskSuggestionsResponse subtasks(Authentication authentication, TaskAssistanceRequest request) {
        requireCredential(authentication);
        requireTaskDraft(request);
        return new SubtaskSuggestionsResponse(List.of("First subtask", "Second subtask"));
    }

    OcrStructureResponse structureOcr(Authentication authentication, OcrStructureRequest request) {
        requireCredential(authentication);
        if (request.rawText() == null || request.rawText().isBlank()) {
            throw new InvalidTaskAssistanceException("OCR text is required.");
        }
        return new OcrStructureResponse("OCR task title", "Structured from OCR text.", List.of());
    }

    private void requireCredential(Authentication authentication) {
        if ("UNAVAILABLE".equals(aiCredentialStatusService.statusFor(authentication).selection())) {
            throw new InvalidTaskAssistanceException("AI credential is unavailable.");
        }
    }

    private void requireTaskDraft(TaskAssistanceRequest request) {
        if (request.title() == null || request.title().isBlank())
```

```
    || request.description() == null || request.description().isBlank()) {  
        throw new InvalidTaskAssistanceException("Task title and description are required.");  
    }  
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/ai/VaultPersonalCredentialStore.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ai/VaultPersonalCredentialStore.java>

```
package com.securefromscratch.beeai;
```

```
import java.io.IOException;
```

```
import java.net.URI;
```

```
import java.net.URLEncoder;
```

```
import java.net.http.HttpClient;
```

```
import java.net.http.HttpRequest;
```

```
import java.net.http.HttpResponse;
```

```
import java.nio.charset.StandardCharsets;
```

```
import java.time.Duration;
```

```
import java.util.Map;
```

```
import java.util.Optional;
```

```
import com.fasterxml.jackson.core.JsonProcessingException;
```

```
import com.fasterxml.jackson.databind.JsonNode;
```

```
import com.fasterxml.jackson.databind.ObjectMapper;
```

```
import org.springframework.beans.factory.annotation.Value;
```

```
import org.springframework.stereotype.Repository;
```

```
@Repository
```

```
class VaultPersonalCredentialStore {
```

```
    private final HttpClient httpClient;
```

```
    private final ObjectMapper objectMapper;
```

```
    private final String vaultUri;
```

```
    private final String vaultToken;
```

```
    private final String mount;
```

```
    VaultPersonalCredentialStore(
```

```
        ObjectMapper objectMapper,
```

```
        @Value("${busybee.secrets.vault.uri:${VAULT_ADDR:http://192.168.48.1:8200}}") String vaultUri,
```

```
        @Value("${busybee.secrets.vault.token:${VAULT_TOKEN:root}}") String vaultToken,
```

```
        @Value("${busybee.secrets.vault.mount:secret}") String mount
```

```
    ){
```

```
        this.httpClient = HttpClient.newBuilder().connectTimeout(Duration.ofSeconds(3)).build();
```

```
        this.objectMapper = objectMapper;
```

```
        this.vaultUri = vaultUri.replaceAll("/+$", "");
```

```
        this.vaultToken = vaultToken;
```

```
        this.mount = mount;
```

```
    }
```

```
    Optional<StoredPersonalCredential> find(String username) {
```

```
        HttpResponse<String> response = send(request("GET", username, HttpRequest.BodyPublishers.noBody()));
```

```

if (response.statusCode() == 404) {
    return Optional.empty();
}
if (response.statusCode() < 200 || response.statusCode() >= 300) {
    throw new AiCredentialStoreException();
}

try {
    JsonNode data = objectMapper.readTree(response.body()).path("data").path("data");
    AiProvider provider = AiProvider.valueOf(data.path("provider").asText());
    String apiKey = data.path("apiKey").asText();
    if (apiKey.isBlank()) {
        throw new AiCredentialStoreException();
    }
    return Optional.of(new StoredPersonalCredential(provider, apiKey));
} catch (JsonProcessingException | IllegalArgumentException exception) {
    throw new AiCredentialStoreException();
}
}

void save(String username, StoredPersonalCredential credential) {
    try {
        String payload = objectMapper.writeValueAsString(Map.of(
            "data", Map.of("provider", credential.provider().name(), "apiKey", credential.apiKey())
        ));
        HttpResponse<String> response = send(request("POST", username, HttpRequest.BodyPublishers.ofString(payload)));
        if (response.statusCode() < 200 || response.statusCode() >= 300) {
            throw new AiCredentialStoreException();
        }
    } catch (JsonProcessingException exception) {
        throw new AiCredentialStoreException();
    }
}

void delete(String username) {
    HttpResponse<String> response = send(request("DELETE", username, HttpRequest.BodyPublishers.noBody()));
    if (response.statusCode() != 404 && (response.statusCode() < 200 || response.statusCode() >= 300)) {
        throw new AiCredentialStoreException();
    }
}

private HttpRequest request(String method, String username, HttpRequest.BodyPublisher body) {
    return HttpRequest.newBuilder(secretUri(username))
        .timeout(Duration.ofSeconds(5))

```

```

        .header("X-Vault-Token", vaultToken)
        .header("Content-Type", "application/json")
        .method(method, body)
        .build();
    }

    private URI secretUri(String username) {
        String encodedUsername = URLEncoder.encode(username, StandardCharsets.UTF_8);
        return URI.create(vaultUri + "/v1/" + mount + "/data/busybee/ai/credentials/" + encodedUsername);
    }

    private HttpResponse<String> send(HttpRequest request) {
        try {
            return httpClient.send(request, HttpResponse.BodyHandlers.ofString());
        } catch (IOException exception) {
            throw new AiCredentialStoreException();
        } catch (InterruptedException exception) {
            Thread.currentThread().interrupt();
            throw new AiCredentialStoreException();
        }
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/comment/AttachmentController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/AttachmentController.java>

```

package com.securefromscratch.busybee.comment;

import java.nio.charset.StandardCharsets;
import java.util.Map;

import org.springframework.http.ContentDisposition;
import org.springframework.http.HttpHeaders;
import org.springframework.http.HttpStatus;
import org.springframework.http.MediaType;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.bind.annotation.RestController;

@RestController
class AttachmentController {

    private final AttachmentRetrievalService attachmentRetrievalService;

    AttachmentController(AttachmentRetrievalService attachmentRetrievalService) {
        this.attachmentRetrievalService = attachmentRetrievalService;
    }

    @GetMapping("/attachment")
    ResponseEntity<byte[]> attachment(@RequestParam("file") String attachmentId) {
        RetrievedAttachment attachment = attachmentRetrievalService.findAttachment(attachmentId);
        ContentDisposition disposition = ContentDisposition.attachment()
            .filename(attachment.originalFilename(), StandardCharsets.UTF_8)
            .build();
        return ResponseEntity.ok()
            .contentType(MediaType.parseMediaType(attachment.contentType()))
            .header(HttpHeaders.CONTENT_DISPOSITION, disposition.toString())
            .body(attachment.content());
    }

    @ExceptionHandler(AttachmentStorageException.class)
    ResponseEntity<Map<String, String>> attachmentError(AttachmentStorageException exception) {
        return ResponseEntity.status(HttpStatus.NOT_FOUND).body(Map.of("error", exception.getMessage()));
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/comment/AttachmentMetadataRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/AttachmentMetadataRepository.java>

```

package com.securefromscratch.busybee.comment;

import java.util.Optional;

import org.jooq.DSLContext;
import org.jooq.impl.DSL;
import org.springframework.stereotype.Repository;

@Repository
class AttachmentMetadataRepository {

    private static final org.jooq.Table<?> STORED_FILES = DSL.table("stored_files");
    private static final org.jooq.Field<String> FILE_ID = DSL.field("file_id", String.class);
    private static final org.jooq.Field<String> ORIGINAL_NAME = DSL.field("original_name", String.class);
    private static final org.jooq.Field<String> CONTENT_TYPE = DSL.field("content_type", String.class);
    private static final org.jooq.Field<String> FILE_KIND = DSL.field("file_kind", String.class);
    private static final org.jooq.Field<String> STORAGE_NAME = DSL.field("storage_name", String.class);
    private static final org.jooq.Field<String> UPLOADED_BY = DSL.field("uploaded_by", String.class);

    private final DSLContext dsl;

    AttachmentMetadataRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    void save(StoredAttachment attachment, String username) {
        dsl.insertInto(STORED_FILES)
            .columns(FILE_ID, ORIGINAL_NAME, CONTENT_TYPE, FILE_KIND, STORAGE_NAME, UPLOADED_BY)
            .values(
                attachment.fileId(),
                attachment.originalFilename(),
                attachment.contentType(),
                "ATTACHMENT",
                attachment.storageName(),
                username
            )
            .execute();
    }

    Optional<StoredAttachment> findAttachment(String fileId) {
        return dsl.select(FILE_ID, ORIGINAL_NAME, CONTENT_TYPE, STORAGE_NAME)
            .from(STORED_FILES)
            .where(FILE_ID.eq(fileId).and(FILE_KIND.eq("ATTACHMENT")))
            .fetchOptional(record -> new StoredAttachment(

```

```
        record.get(FILE_ID),
        record.get(ORIGINAL_NAME),
        record.get(CONTENT_TYPE),
        record.get(STORAGE_NAME)
    ));
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/AttachmentRetrievalService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/AttachmentRetrievalService.java>

```

package com.securefromscratch.busybee.comment;

import java.util.UUID;

import org.springframework.stereotype.Service;

@Service
class AttachmentRetrievalService {

    private final AttachmentMetadataRepository attachmentMetadataRepository;
    private final AttachmentStorageService attachmentStorageService;

    AttachmentRetrievalService(
        AttachmentMetadataRepository attachmentMetadataRepository,
        AttachmentStorageService attachmentStorageService
    ){
        this.attachmentMetadataRepository = attachmentMetadataRepository;
        this.attachmentStorageService = attachmentStorageService;
    }

    RetrievedAttachment findAttachment(String attachmentId) {
        String validAttachmentId = validAttachmentId(attachmentId);
        StoredAttachment attachment = attachmentMetadataRepository.findAttachment(validAttachmentId)
            .orElseThrow(() -> new AttachmentStorageException("Attachment was not found."));
        return new RetrievedAttachment(
            attachment.originalFilename(),
            attachment.contentType(),
            attachmentStorageService.load(attachment)
        );
    }

    private String validAttachmentId(String attachmentId) {
        if (attachmentId == null) {
            throw new AttachmentStorageException("Attachment was not found.");
        }

        try {
            return UUID.fromString(attachmentId).toString();
        } catch (IllegalArgumentException exception) {
            throw new AttachmentStorageException("Attachment was not found.");
        }
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/comment/AttachmentStorageException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/AttachmentStorageException.java>

```
package com.securefromscratch.busybee.comment;

class AttachmentStorageException extends RuntimeException {

    AttachmentStorageException(String message) {
        super(message);
    }

    AttachmentStorageException(String message, Throwable cause) {
        super(message, cause);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/AttachmentStorageService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/AttachmentStorageService.java>

```

package com.securefromscratch.busybee.comment;

import java.io.IOException;
import java.nio.file.Files;
import java.nio.file.Path;
import java.nio.file.StandardOpenOption;
import java.util.Locale;
import java.util.Map;
import java.util.UUID;

import org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Service;

@Service
class AttachmentStorageService {

    private static final Map<String, String> ATTACHMENT_CONTENT_TYPES = Map.ofEntries(
        Map.entry("txt", "text/plain"),
        Map.entry("csv", "text/csv"),
        Map.entry("pdf", "application/pdf"),
        Map.entry("doc", "application/msword"),
        Map.entry("docx", "application/vnd.openxmlformats-officedocument.wordprocessingml.document"),
        Map.entry("xls", "application/vnd.ms-excel"),
        Map.entry("xlsx", "application/vnd.openxmlformats-officedocument.spreadsheetml.sheet"),
        Map.entry("json", "application/json"),
        Map.entry("zip", "application/zip")
    );

    private final Path storageDirectory;

    AttachmentStorageService(@Value("${busybee.file-storage.directory}") String storageDirectory) {
        this.storageDirectory = Path.of(storageDirectory).toAbsolutePath().normalize();
    }

    StoredAttachment store(AttachmentUpload upload) {
        String originalFilename = validFilename(upload.originalFilename());
        String contentType = ATTACHMENT_CONTENT_TYPES.get(extensionOf(originalFilename));
        if (contentType == null) {
            throw new InvalidCommentException("Unsupported attachment file type.");
        }
        if (upload.content().length == 0) {
            throw new InvalidCommentException("Attachment file is empty.");
        }
    }
}

```

```

String fileId = UUID.randomUUID().toString();
String storageName = fileId + ".bin";
Path storagePath = storageDirectory.resolve(storageName).normalize();
if (!storagePath.startsWith(storageDirectory)) {
    throw new InvalidCommentException("Attachment filename is invalid.");
}

try {
    Files.createDirectories(storageDirectory);
    Files.write(storagePath, upload.content(), StandardOpenOption.CREATE_NEW);
} catch (IOException exception) {
    throw new AttachmentStorageException("Attachment could not be stored.", exception);
}

return new StoredAttachment(fileId, originalFilename, contentType, storageName);
}

byte[] load(StoredAttachment attachment) {
    Path storagePath = storageDirectory.resolve(attachment.storageName()).normalize();
    if (!storagePath.startsWith(storageDirectory)) {
        throw new AttachmentStorageException("Attachment could not be loaded.");
    }

    try {
        return Files.readAllBytes(storagePath);
    } catch (IOException exception) {
        throw new AttachmentStorageException("Attachment could not be loaded.", exception);
    }
}

private String validFilename(String filename) {
    if (filename == null || filename.isBlank() || filename.contains("/") || filename.contains("\\")) {
        throw new InvalidCommentException("Attachment filename is invalid.");
    }

    return filename;
}

private String extensionOf(String filename) {
    int lastDot = filename.lastIndexOf('.');
    if (lastDot <= 0 || lastDot == filename.length() - 1) {
        throw new InvalidCommentException("Unsupported attachment file type.");
    }
}

```

```
    return filename.substring(lastDot + 1).toLowerCase(Locale.ROOT);  
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/AttachmentUpload.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/AttachmentUpload.java>

```
package com.securefromscratch.busybee.comment;  
  
record AttachmentUpload(String originalFilename, byte[] content) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentCreateRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentCreateRequest.java>

```
package com.securefromscratch.busybee.comment;  
  
record CommentCreateRequest(String text, String taskid, String commentid) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentCreateResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentCreateResponse.java>

```
package com.securefromscratch.busybee.comment;

record CommentCreateResponse(String commentid) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentCreationController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentCreationController.java>

```

package com.securefromscratch.busybee.comment;

import java.io.IOException;
import java.util.Map;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestPart;
import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.multipart.MultipartFile;

@RestController
class CommentCreationController {

    private final CommentCreationService commentCreationService;

    CommentCreationController(CommentCreationService commentCreationService) {
        this.commentCreationService = commentCreationService;
    }

    @PostMapping(path = "/comment", consumes = "multipart/form-data")
    CommentCreateResponse createComment(
        Authentication authentication,
        @RequestPart("commentFields") CommentCreateRequest request,
        @RequestPart(value = "file", required = false) MultipartFile file
    ){
        return commentCreationService.createComment(authentication.getName(), request, imageUpload(file));
    }

    @ExceptionHandler(InvalidCommentException.class)
    ResponseEntity<Map<String, String>> invalidComment(InvalidCommentException exception) {
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
    }

    private ImageUpload imageUpload(MultipartFile file) {
        if (file == null) {
            return null;
        }

        try {
            return new ImageUpload(file.getOriginalFilename(), file.getBytes());
        }
    }
}

```

```
    } catch (IOException exception) {  
        throw new InvalidCommentException("Image file could not be read.");  
    }  
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentCreationService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentCreationService.java>

```

package com.securefromscratch.busybee.comment;

import java.util.UUID;

import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
class CommentCreationService {

    private final CommentRepository commentRepository;
    private final ImageStorageService imageStorageService;
    private final ImageMetadataRepository imageMetadataRepository;
    private final AttachmentStorageService attachmentStorageService;
    private final AttachmentMetadataRepository attachmentMetadataRepository;

    CommentCreationService(
        CommentRepository commentRepository,
        ImageStorageService imageStorageService,
        ImageMetadataRepository imageMetadataRepository,
        AttachmentStorageService attachmentStorageService,
        AttachmentMetadataRepository attachmentMetadataRepository
    ){
        this.commentRepository = commentRepository;
        this.imageStorageService = imageStorageService;
        this.imageMetadataRepository = imageMetadataRepository;
        this.attachmentStorageService = attachmentStorageService;
        this.attachmentMetadataRepository = attachmentMetadataRepository;
    }

    @Transactional
    CommentCreateResponse createComment(String username, CommentCreateRequest request, ImageUpload image)
    {
        String taskId = validTaskId(request.taskId());
        if (request.text() == null || request.text().isBlank()) {
            throw new InvalidCommentException("Comment text is required.");
        }
        if (!commentRepository.taskExists(taskId)) {
            throw new InvalidCommentException("Task was not found.");
        }

        CommentParent parent = parentComment(request.commentid(), taskId);

        String commentId = UUID.randomUUID().toString();
        StoredImage image = imageUpload == null || !isImageUpload(imageUpload)

```

```

        ? null
        : imageStorageService.store(imageUpload);
StoredAttachment attachment = imageUpload == null || image != null
    ? null
    : attachmentStorageService.store(new AttachmentUpload(imageUpload.originalFilename(), imageUpload.contentType()));
if (image != null) {
    imageMetadataRepository.save(image, username);
}
if (attachment != null) {
    attachmentMetadataRepository.save(attachment, username);
}
commentRepository.create(
    commentId,
    taskId,
    parent == null ? null : parent.commentId(),
    request.text(),
    image == null ? null : image.fileId(),
    attachment == null ? null : attachment.fileId(),
    parent == null ? 0 : parent.indent() + 1,
    username
);
return new CommentCreateResponse(commentId);
}

private String validTaskId(String taskId) {
    if (taskId == null) {
        throw new InvalidCommentException("Task ID is invalid.");
    }

    try {
        return UUID.fromString(taskId).toString();
    } catch (IllegalArgumentException exception) {
        throw new InvalidCommentException("Task ID is invalid.");
    }
}

private CommentParent parentComment(String parentCommentId, String taskId) {
    if (parentCommentId == null) {
        return null;
    }

    String validParentCommentId = validCommentId(parentCommentId);
    CommentParent parent = commentRepository.findParent(validParentCommentId)
        .orElseThrow(() -> new InvalidCommentException("Parent comment was not found."));

```

```

    if (!taskId.equals(parent.taskId())) {
        throw new InvalidCommentException("Parent comment belongs to another task.");
    }
    return parent;
}

private String validCommentId(String commentId) {
    try {
        return UUID.fromString(commentId).toString();
    } catch (IllegalArgumentException exception) {
        throw new InvalidCommentException("Parent comment ID is invalid.");
    }
}

private boolean isImageUpload(ImageUpload upload) {
    String filename = upload.originalFilename();
    if (filename == null) {
        return false;
    }

    String lowercaseFilename = filename.toLowerCase(java.util.Locale.ROOT);
    return lowercaseFilename.endsWith(".png")
        || lowercaseFilename.endsWith(".jpg")
        || lowercaseFilename.endsWith(".jpeg")
        || lowercaseFilename.endsWith(".gif")
        || lowercaseFilename.endsWith(".webp")
        || lowercaseFilename.endsWith(".svg");
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentListResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentListResponse.java>

```
package com.securefromscratch.busybee.comment;

import java.time.LocalDateTime;
import java.util.List;

import com.securefromscratch.busybee.preview.TaskLinkPreviewResponse;

public record CommentListResponse(
    String commentid,
    String text,
    String image,
    String imageFilename,
    String attachment,
    String attachmentFilename,
    int indent,
    String createdBy,
    LocalDateTime createdOn,
    List<TaskLinkPreviewResponse> linkPreviews
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentListingService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentListingService.java>

```

package com.securefromscratch.busybee.comment;

import java.util.List;

import com.securefromscratch.busybee.preview.CommentPreviewListingService;

import org.springframework.stereotype.Service;

@Service
public class CommentListingService {

    private final CommentRepository commentRepository;
    private final ImageMetadataRepository imageMetadataRepository;
    private final AttachmentMetadataRepository attachmentMetadataRepository;
    private final CommentPreviewListingService commentPreviewListingService;

    CommentListingService(
        CommentRepository commentRepository,
        ImageMetadataRepository imageMetadataRepository,
        AttachmentMetadataRepository attachmentMetadataRepository,
        CommentPreviewListingService commentPreviewListingService
    ){
        this.commentRepository = commentRepository;
        this.imageMetadataRepository = imageMetadataRepository;
        this.attachmentMetadataRepository = attachmentMetadataRepository;
        this.commentPreviewListingService = commentPreviewListingService;
    }

    public List<CommentListResponse> listComments(String taskId) {
        return commentRepository.findByTaskId(taskId).stream()
            .map(comment -> commentResponse(comment))
            .toList();
    }

    private CommentListResponse commentResponse(StoredComment comment) {
        StoredImage image = comment.imageId() == null
            ? null
            : imageMetadataRepository.findImage(comment.imageId()).orElse(null);
        StoredAttachment attachment = comment.attachmentId() == null
            ? null
            : attachmentMetadataRepository.findAttachment(comment.attachmentId()).orElse(null);
        return new CommentListResponse(
            comment.commentId(),
            comment.text(),

```

```

        comment.imageId(),
        image == null ? null : image.originalFilename(),
        comment.attachmentId(),
        attachment == null ? null : attachment.originalFilename(),
        comment.indent(),
        comment.createdBy(),
        comment.createdAt(),
        commentPreviewListingService.listCommentPreviews(comment.commentId())
    );
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentParent.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentParent.java>

```

package com.securefromscratch.busybee.comment;

record CommentParent(String commentId, String taskId, int indent) {
}

```

busybee/src/main/java/com/securefromscratch/busybee/comment/CommentRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/CommentRepository.java>

```

package com.securefromscratch.busybee.comment;

import java.time.LocalDateTime;
import java.util.List;

import org.jooq.DSLContext;
import org.jooq.impl.DSL;
import org.springframework.stereotype.Repository;

@Repository
class CommentRepository {

    private static final org.jooq.Table<?> TASKS = DSL.table("tasks");
    private static final org.jooq.Field<String> TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Table<?> COMMENTS = DSL.table("comments");
    private static final org.jooq.Field<String> COMMENT_ID = DSL.field("comment_id", String.class);
    private static final org.jooq.Field<String> COMMENT_TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Field<String> PARENT_COMMENT_ID = DSL.field("parent_comment_id", String.class);
    private static final org.jooq.Field<String> TEXT = DSL.field("text", String.class);
    private static final org.jooq.Field<String> IMAGE_FILE_ID = DSL.field("image_file_id", String.class);
    private static final org.jooq.Field<String> ATTACHMENT_FILE_ID = DSL.field("attachment_file_id", String.class);
    private static final org.jooq.Field<Integer> INDENT = DSL.field("indent", Integer.class);
    private static final org.jooq.Field<String> CREATED_BY = DSL.field("created_by", String.class);
    private static final org.jooq.Field<LocalDateTime> CREATED_AT = DSL.field("created_at", LocalDateTime.class);

    private final DSLContext dsl;

    CommentRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    boolean taskExists(String taskId) {
        return dsl.fetchExists(dsl.selectOne().from(TASKS).where(TASK_ID.eq(taskId)));
    }

    void create(
        String commentId,
        String taskId,
        String parentCommentId,
        String text,
        String imageId,
        String attachmentId,
        int indent,
        String username
    ) {

```

```

){
    dsl.insertInto(COMMENTS)
        .columns(
            COMMENT_ID,
            COMMENT_TASK_ID,
            PARENT_COMMENT_ID,
            TEXT,
            IMAGE_FILE_ID,
            ATTACHMENT_FILE_ID,
            INDENT,
            CREATED_BY
        )
        .values(commentId, taskId, parentCommentId, text, imageId, attachmentId, indent, username)
        .execute();
}

```

```

java.util.Optional<CommentParent> findParent(String commentId) {
    return dsl.select(COMMENT_ID, COMMENT_TASK_ID, INDENT)
        .from(COMMENTS)
        .where(COMMENT_ID.eq(commentId))
        .fetchOptional(record -> new CommentParent(
            record.get(COMMENT_ID),
            record.get(COMMENT_TASK_ID),
            record.get(INDENT)
        ));
}

```

```

List<StoredComment> findByTaskId(String taskId) {
    return dsl.select(COMMENT_ID, TEXT, IMAGE_FILE_ID, ATTACHMENT_FILE_ID, INDENT, CREATED_BY, CREATED_AT)
        .from(COMMENTS)
        .where(COMMENT_TASK_ID.eq(taskId))
        .orderBy(CREATED_AT.asc(), COMMENT_ID.asc())
        .fetch(record -> new StoredComment(
            record.get(COMMENT_ID),
            record.get(TEXT),
            record.get(IMAGE_FILE_ID),
            record.get(ATTACHMENT_FILE_ID),
            record.get(INDENT),
            record.get(CREATED_BY),
            record.get(CREATED_AT)
        ));
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/comment/ImageController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/ImageController.java>

```
package com.securefromscratch.busybee.comment;

import java.util.Map;

import org.springframework.http.HttpStatus;
import org.springframework.http.MediaType;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.bind.annotation.RestController;

@RestController
class ImageController {

    private final ImageRetrievalService imageRetrievalService;

    ImageController(ImageRetrievalService imageRetrievalService) {
        this.imageRetrievalService = imageRetrievalService;
    }

    @GetMapping("/image")
    ResponseEntity<byte[]> image(@RequestParam("img") String imageUrl) {
        RetrievedImage image = imageRetrievalService.findImage(imageId);
        return ResponseEntity.ok().contentType(MediaType.parseMediaType(image.contentType)).body(image.content());
    }

    @ExceptionHandler(ImageStorageException.class)
    ResponseEntity<Map<String, String>> imageError(ImageStorageException exception) {
        return ResponseEntity.status(HttpStatus.NOT_FOUND).body(Map.of("error", exception.getMessage()));
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/ImageMetadataRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/ImageMetadataRepository.java>

```

package com.securefromscratch.busybee.comment;

import java.util.Optional;

import org.jooq.DSLContext;
import org.jooq.impl.DSL;
import org.springframework.stereotype.Repository;

@Repository
public class ImageMetadataRepository {

    private static final org.jooq.Table<?> STORED_FILES = DSL.table("stored_files");
    private static final org.jooq.Field<String> FILE_ID = DSL.field("file_id", String.class);
    private static final org.jooq.Field<String> ORIGINAL_NAME = DSL.field("original_name", String.class);
    private static final org.jooq.Field<String> CONTENT_TYPE = DSL.field("content_type", String.class);
    private static final org.jooq.Field<String> FILE_KIND = DSL.field("file_kind", String.class);
    private static final org.jooq.Field<String> STORAGE_NAME = DSL.field("storage_name", String.class);
    private static final org.jooq.Field<String> UPLOADED_BY = DSL.field("uploaded_by", String.class);

    private final DSLContext dsl;

    ImageMetadataRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    void save(StoredImage image, String username) {
        dsl.insertInto(STORED_FILES)
            .columns(FILE_ID, ORIGINAL_NAME, CONTENT_TYPE, FILE_KIND, STORAGE_NAME, UPLOADED_BY)
            .values(image.fileId(), image.originalFilename(), image.contentType(), "IMAGE", image.storageName(), username)
            .execute();
    }

    public Optional<StoredImage> findImage(String fileId) {
        return dsl.select(FILE_ID, ORIGINAL_NAME, CONTENT_TYPE, STORAGE_NAME)
            .from(STORED_FILES)
            .where(FILE_ID.eq(fileId).and(FILE_KIND.eq("IMAGE")))
            .fetchOptional(record -> new StoredImage(
                record.get(FILE_ID),
                record.get(ORIGINAL_NAME),
                record.get(CONTENT_TYPE),
                record.get(STORAGE_NAME)
            ));
    }
}

```

```
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/ImageViewRetrievalService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/ImageViewRetrievalService.java>

```

package com.securefromscratch.busybee.comment;

import java.util.UUID;

import org.springframework.stereotype.Service;

@Service
class ImageRetrievalService {

    private final ImageMetadataRepository imageMetadataRepository;
    private final ImageStorageService imageStorageService;

    ImageRetrievalService(ImageMetadataRepository imageMetadataRepository, ImageStorageService imageStorageService) {
        this.imageMetadataRepository = imageMetadataRepository;
        this.imageStorageService = imageStorageService;
    }

    RetrievedImage findImage(String imageId) {
        String validImageId = validImageId(imageId);
        StoredImage image = imageMetadataRepository.findImage(validImageId)
            .orElseThrow(() -> new ImageStorageException("Image was not found."));
        return new RetrievedImage(image.contentType(), imageStorageService.load(image));
    }

    private String validImageId(String imageId) {
        if (imageId == null) {
            throw new ImageStorageException("Image was not found.");
        }

        try {
            return UUID.fromString(imageId).toString();
        } catch (IllegalArgumentException exception) {
            throw new ImageStorageException("Image was not found.");
        }
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/comment/ImageStorageException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1

271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/ImageStorageException.java>

```
package com.securefromscratch.busybee.comment;

class ImageStorageException extends RuntimeException {

    ImageStorageException(String message) {
        super(message);
    }

    ImageStorageException(String message, Throwable cause) {
        super(message, cause);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/ImageStorageService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/ImageStorageService.java>

```

package com.securefromscratch.busybee.comment;

import java.io.IOException;
import java.nio.file.Files;
import java.nio.file.Path;
import java.nio.file.StandardOpenOption;
import java.util.Locale;
import java.util.Map;
import java.util.UUID;

import org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Service;

@Service
public class ImageStorageService {

    private static final Map<String, String> IMAGE_CONTENT_TYPES = Map.of(
        "png", "image/png",
        "jpg", "image/jpeg",
        "jpeg", "image/jpeg",
        "gif", "image/gif",
        "webp", "image/webp",
        "svg", "image/svg+xml"
    );

    private final Path storageDirectory;

    ImageStorageService(@Value("${busybee.file-storage.directory}") String storageDirectory) {
        this.storageDirectory = Path.of(storageDirectory).toAbsolutePath().normalize();
    }

    StoredImage store(ImageUpload upload) {
        String originalFilename = validFilename(upload.originalFilename());
        String extension = extensionOf(originalFilename);
        String contentType = IMAGE_CONTENT_TYPES.get(extension);
        if (contentType == null) {
            throw new InvalidCommentException("Unsupported image file type.");
        }
        if (upload.content().length == 0) {
            throw new InvalidCommentException("Image file is empty.");
        }

        String fileId = UUID.randomUUID().toString();
        String storageName = fileId + ".bin";
    }

```

```

Path storagePath = storageDirectory.resolve(storageName).normalize();
if (!storagePath.startsWith(storageDirectory)) {
    throw new InvalidCommentException("Image filename is invalid.");
}

try {
    Files.createDirectories(storageDirectory);
    Files.write(storagePath, upload.content(), StandardOpenOption.CREATE_NEW);
} catch (IOException exception) {
    throw new ImageStorageException("Image could not be stored.", exception);
}

return new StoredImage(fileId, originalFilename, contentType, storageName);
}

byte[] load(StoredImage image) {
    Path storagePath = pathFor(image);

    try {
        return Files.readAllBytes(storagePath);
    } catch (IOException exception) {
        throw new ImageStorageException("Image could not be loaded.", exception);
    }
}

public Path pathFor(StoredImage image) {
    Path storagePath = storageDirectory.resolve(image.storageName()).normalize();
    if (!storagePath.startsWith(storageDirectory)) {
        throw new ImageStorageException("Image could not be loaded.");
    }
    return storagePath;
}

private String validFilename(String filename) {
    if (filename == null || filename.isBlank() || filename.contains("/") || filename.contains("\\")) {
        throw new InvalidCommentException("Image filename is invalid.");
    }

    return filename;
}

private String extensionOf(String filename) {
    int lastDot = filename.lastIndexOf('.');
    if (lastDot <= 0 || lastDot == filename.length() - 1) {

```

```
        throw new InvalidCommentException("Unsupported image file type.");
    }

    return filename.substring(lastDot + 1).toLowerCase(Locale.ROOT);
}
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/ImageUpload.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/ImageUpload.java>

```
package com.securefromscratch.busybee.comment;

record ImageUpload(String originalFilename, byte[] content) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/InvalidCommentException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/InvalidCommentException.java>

```
package com.securefromscratch.busybee.comment;

class InvalidCommentException extends RuntimeException {

    InvalidCommentException(String message) {
        super(message);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/RetrievedAttachment.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/RetrievedAttachment.java>

```
package com.securefromscratch.busybee.comment;
```

```
record RetrievedAttachment(String originalFilename, String contentType, byte[] content) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/RetrievedImage.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/RetrievedImage.java>

```
package com.securefromscratch.busybee.comment;
```

```
record RetrievedImage(String contentType, byte[] content) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/StoredAttachment.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/StoredAttachment.java>

```
package com.securefromscratch.busybee.comment;
```

```
record StoredAttachment(String fileId, String originalFilename, String contentType, String storageName) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/StoredComment.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1>

271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/StoredComment.java>

```
package com.securefromscratch.busybee.comment;

import java.time.LocalDateTime;

record StoredComment(
    String commentId,
    String text,
    String imageId,
    String attachmentId,
    int indent,
    String createdBy,
    LocalDateTime createdAt
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/comment/StoredImage.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/comment/StoredImage.java>

```
package com.securefromscratch.busybee.comment;

public record StoredImage(String fileId, String originalFilename, String contentType, String storageName) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/config/JsonConfiguration.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/config/JsonConfiguration.java>

```
package com.securefromscratch.busybee.config;

import com.fasterxml.jackson.databind.ObjectMapper;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;

@Configuration
public class JsonConfiguration {

    @Bean
    ObjectMapper objectMapper() {
        return new ObjectMapper().findAndRegisterModules();
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/config/SecurityConfiguration.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/config/SecurityConfiguration.java>

```

package com.securefromscratch.busybee.config;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.boot.context.properties.EnableConfigurationProperties;
import org.springframework.security.config.annotation.web.builders.HttpSecurity;
import org.springframework.security.crypto.bcrypt.BCryptPasswordEncoder;
import org.springframework.security.crypto.password.PasswordEncoder;
import org.springframework.security.web.authentication.AnonymousAuthenticationFilter;
import org.springframework.security.web.SecurityFilterChain;
import org.springframework.security.web.csrf.CookieCsrfTokenRepository;

import com.securefromscratch.busybee.session.DemoSessionAuthenticationFilter;
import com.securefromscratch.busybee.session.DemoSessionProperties;

@Configuration
@EnableConfigurationProperties(DemoSessionProperties.class)
public class SecurityConfiguration {

    @Bean
    SecurityFilterChain securityFilterChain(
        HttpSecurity http,
        DemoSessionAuthenticationFilter demoSessionAuthenticationFilter
    ) throws Exception {
        return http
            .csrf(csrf -> csrf.csrfTokenRepository(CookieCsrfTokenRepository.withHttpOnlyFalse()))
            .authorizeHttpRequests(authorize -> authorize
                .requestMatchers("/", "/index.html", "/register", "/gencsrfToken", "/register.js", "/welcome.css").permitAll()
                .requestMatchers("/admin/**").hasRole("ADMIN")
                .anyRequest().authenticated()
            )
            .formLogin(form -> form.defaultSuccessUrl("/main/main.html", true))
            .logout(logout -> logout.logoutSuccessUrl("/index.html"))
            .addFilterBefore(demoSessionAuthenticationFilter, AnonymousAuthenticationFilter.class)
            .build();
    }

    @Bean
    PasswordEncoder passwordEncoder() {
        return new BCryptPasswordEncoder();
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/extra/ExportedTask.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/extra/ExportedTask.java>

```
package com.securefromscratch.busybee.extra;
```

```
import java.time.LocalDate;
```

```
import java.time.LocalTime;
```

```
import java.util.List;
```

```
record ExportedTask(
```

```
    String name,
```

```
    String desc,
```

```
    LocalDate dueDate,
```

```
    LocalTime dueTime,
```

```
    List<String> responsibilityOf
```

```
){
```

```
}
```

busybee/src/main/java/com/securefromscratch/busybee/extra/ImportedTask.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/extra/ImportedTask.java>

```
package com.securefromscratch.busybee.extra;
```

```
import java.time.LocalDate;
```

```
import java.time.LocalTime;
```

```
import java.util.List;
```

```
public record ImportedTask(
```

```
    String name,
```

```
    String desc,
```

```
    LocalDate dueDate,
```

```
    LocalTime dueTime,
```

```
    List<String> responsibilityOf
```

```
) {
```

```
}
```

busybee/src/main/java/com/securefromscratch/busybee/extra/InvalidTaskTransferException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/extra/InvalidTaskTransferException.java>

```
package com.securefromscratch.busybee.extra;
```

```
class InvalidTaskTransferException extends RuntimeException {
```

```
    InvalidTaskTransferException(String message) {
```

```
        super(message);
```

```
    }
```

```
}
```

busybee/src/main/java/com/securefromscratch/busybee/extra/TaskTransferController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/extra/TaskTransferController.java>

```

package com.securefromscratch.busybee.extra;

import java.io.IOException;
import java.util.List;
import java.util.Map;

import com.fasterxml.jackson.core.type.TypeReference;
import com.fasterxml.jackson.databind.ObjectMapper;

import org.springframework.http.ContentDisposition;
import org.springframework.http.HttpHeaders;
import org.springframework.http.HttpStatus;
import org.springframework.http.MediaType;
import org.springframework.http.ResponseEntity;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestPart;
import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.multipart.MultipartFile;

@RestController
class TaskTransferController {

    private final TaskTransferService taskTransferService;
    private final ObjectMapper objectMapper;

    TaskTransferController(TaskTransferService taskTransferService, ObjectMapper objectMapper) {
        this.taskTransferService = taskTransferService;
        this.objectMapper = objectMapper;
    }

    @GetMapping("/extra/export")
    ResponseEntity<byte[]> exportTasks(Authentication authentication) throws IOException {
        byte[] contents = objectMapper.writeValueAsBytes(taskTransferService.exportTasks(authentication));
        return ResponseEntity.ok()
            .contentType(MediaType.APPLICATION_OCTET_STREAM)
            .header(HttpHeaders.CONTENT_DISPOSITION, ContentDisposition.attachment().filename("tasks.ser").build().toSingleString())
            .body(contents);
    }

    @PostMapping(path = "/extra/import", consumes = MediaType.MULTIPART_FORM_DATA_VALUE)
    ResponseEntity<Void> importTasks(Authentication authentication, @RequestPart("file") MultipartFile file) {

```

```

try {
    List<ImportedTask> tasks = objectMapper.readValue(file.getBytes(), new TypeReference<>() { });
    taskTransferService.importTasks(authentication, tasks);
    return ResponseEntity.ok().build();
} catch (IOException exception) {
    throw new InvalidTaskTransferException("Import file is invalid.");
}
}

@ExceptionHandler(InvalidTaskTransferException.class)
ResponseEntity<Map<String, String>> invalidImport(InvalidTaskTransferException exception) {
    return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/extra/TaskTransferRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/extra/TaskTransferRepository.java>

```

package com.securefromscratch.busybee.extra;

import java.time.LocalDate;
import java.time.LocalTime;
import java.util.List;
import java.util.UUID;

import org.jooq.DSLContext;
import org.jooq.impl.DSL;
import org.springframework.stereotype.Repository;

@Repository
class TaskTransferRepository {

    private static final org.jooq.Table<?> TASKS = DSL.table("tasks");
    private static final org.jooq.Field<String> TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Field<String> NAME = DSL.field("name", String.class);
    private static final org.jooq.Field<String> DESCRIPTION = DSL.field("description", String.class);
    private static final org.jooq.Field<LocalDate> DUE_DATE = DSL.field("due_date", LocalDate.class);
    private static final org.jooq.Field<LocalTime> DUE_TIME = DSL.field("due_time", LocalTime.class);
    private static final org.jooq.Field<String> CREATED_BY = DSL.field("created_by", String.class);
    private static final org.jooq.Table<?> TASK_RESPONSIBILITIES = DSL.table("task_responsibilities");
    private static final org.jooq.Field<String> RESPONSIBILITY_TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Field<String> RESPONSIBLE_NAME = DSL.field("responsible_name", String.class);
    private static final org.jooq.Field<Integer> POSITION_INDEX = DSL.field("position_index", Integer.class);

    private final DSLContext dsl;

    TaskTransferRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    List<ExportedTask> findExportedTasks() {
        return dsl.select(TASK_ID, NAME, DESCRIPTION, DUE_DATE, DUE_TIME)
            .from(TASKS)
            .orderBy(TASK_ID.asc())
            .fetch(record -> new ExportedTask(
                record.get(NAME),
                record.get(DESCRIPTION),
                record.get(DUE_DATE),
                record.get(DUE_TIME),
                responsibilities(record.get(TASK_ID))
            ));
    }
}

```

```

void createImportedTask(String username, ImportedTask task) {
    String taskId = UUID.randomUUID().toString();
    LocalTime dueTime = task.dueDate() == null ? null : task.dueTime();
    List<String> responsibilities = normalizedResponsibilities(task.responsibilityOf());
    dsl.insertInto(TASKS)
        .columns(TASK_ID, NAME, DESCRIPTION, DUE_DATE, DUE_TIME, CREATED_BY)
        .values(taskId, task.name().trim(), task.desc() == null ? "" : task.desc(), task.dueDate(), dueTime, username)
        .execute();

    for (int index = 0; index < responsibilities.size(); index++) {
        dsl.insertInto(TASK_RESPONSIBILITIES)
            .columns(RESPONSIBILITY_TASK_ID, RESPONSIBLE_NAME, POSITION_INDEX)
            .values(taskId, responsibilities.get(index), index)
            .execute();
    }
}

private List<String> responsibilities(String taskId) {
    return dsl.select(RESPONSIBLE_NAME)
        .from(TASK_RESPONSIBILITIES)
        .where(RESPONSIBILITY_TASK_ID.eq(taskId))
        .orderBy(POSITION_INDEX.asc())
        .fetch(RESPONSIBLE_NAME);
}

private List<String> normalizedResponsibilities(List<String> responsibilities) {
    return responsibilities == null
        ? List.of()
        : responsibilities.stream().filter(value -> value != null).map(String::trim)
            .filter(value -> !value.isEmpty()).toList();
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/extra/TaskTransferService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/extra/TaskTransferService.java>

```

package com.securefromscratch.busybee.extra;

import java.util.List;

import org.springframework.security.access.AccessDeniedException;
import org.springframework.security.core.Authentication;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
class TaskTransferService {

    private final TaskTransferRepository taskTransferRepository;

    TaskTransferService(TaskTransferRepository taskTransferRepository) {
        this.taskTransferRepository = taskTransferRepository;
    }

    List<ExportedTask> exportTasks(Authentication authentication) {
        requireEntitlement(authentication, "EXPORT_ENABLED");
        return taskTransferRepository.findExportedTasks();
    }

    @Transactional
    void importTasks(Authentication authentication, List<ImportedTask> tasks) {
        requireEntitlement(authentication, "IMPORT_ENABLED");
        if (tasks == null) {
            throw new InvalidTaskTransferException("Import file must contain a task list.");
        }
        for (ImportedTask task : tasks) {
            if (task == null || task.name() == null || task.name().isBlank()) {
                throw new InvalidTaskTransferException("Each imported task requires a name.");
            }
            taskTransferRepository.createImportedTask(authentication.getName(), task);
        }
    }

    private void requireEntitlement(Authentication authentication, String entitlement) {
        boolean allowed = authentication.getAuthorities().stream()
            .map(authority -> authority.getAuthority())
            .anyMatch(authority -> authority.equals("ROLE_ADMIN") || authority.equals(entitlement));
        if (!allowed) {
            throw new AccessDeniedException("Required entitlement is missing.");
        }
    }
}

```

```
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrController.java>

```
package com.securefromscratch.busybee.ocr;  
  
import java.util.Map;  
  
import org.springframework.http.HttpStatus;  
import org.springframework.http.ResponseEntity;  
import org.springframework.security.core.Authentication;  
import org.springframework.web.bind.annotation.ExceptionHandler;  
import org.springframework.web.bind.annotation.PostMapping;  
import org.springframework.web.bind.annotation.RequestBody;  
import org.springframework.web.bind.annotation.RestController;  
  
@RestController  
class OcrController {  
  
    private final OcrService ocrService;  
  
    OcrController(OcrService ocrService) {  
        this.ocrService = ocrService;  
    }  
  
    @PostMapping("/ocr/image")  
    OcrDraft extractImage(Authentication authentication, @RequestBody OcrRequest request) {  
        return ocrService.extractImage(authentication, request);  
    }  
  
    @ExceptionHandler(OcrException.class)  
    ResponseEntity<Map<String, String>> ocrError(OcrException exception) {  
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));  
    }  
}
```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrDraft.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrDraft.java>

```
package com.securefromscratch.busybee.ocr;

record OcrDraft(String title, String description, String rawText) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrException.java>

```
package com.securefromscratch.busybee.ocr;

class OcrException extends RuntimeException {

    OcrException(String message) {
        super(message);
    }

    OcrException(String message, Throwable cause) {
        super(message, cause);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrRequest.java>

```
package com.securefromscratch.busybee.ocr;
```

```
record OcrRequest(String image, String language, Integer textLayout) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/ocr/OcrService.java>

```

package com.securefromscratch.busybee.ocr;

import java.io.IOException;
import java.nio.charset.StandardCharsets;
import java.nio.file.Path;
import java.util.List;
import java.util.Set;
import java.util.UUID;
import java.util.concurrent.TimeUnit;

import com.securefromscratch.busybee.comment.ImageMetadataRepository;
import com.securefromscratch.busybee.comment.ImageStorageService;
import com.securefromscratch.busybee.comment.StoredImage;

import org.springframework.beans.factory.annotation.Value;
import org.springframework.security.core.Authentication;
import org.springframework.stereotype.Service;

@Service
class OcrService {

    private static final Set<String> LANGUAGES = Set.of("eng", "heb", "eng+heb");
    private static final Set<Integer> TEXT_LAYOUTS = Set.of(3, 4, 6, 7, 8, 11, 12, 13);

    private final ImageMetadataRepository imageMetadataRepository;
    private final ImageStorageService imageStorageService;
    private final String ocrCommand;

    OcrService(
        ImageMetadataRepository imageMetadataRepository,
        ImageStorageService imageStorageService,
        @Value("${busybee.ocr.command}") String ocrCommand
    ) {
        this.imageMetadataRepository = imageMetadataRepository;
        this.imageStorageService = imageStorageService;
        this.ocrCommand = ocrCommand;
    }

    OcrDraft extractImage(Authentication authentication, OcrRequest request) {
        requireOcrAccess(authentication);
        String imageId = validImageId(request.image());
        validateParameters(request);
        StoredImage image = imageMetadataRepository.findImage(imageId)
            .orElseThrow(() -> new OcrException("Image was not found."));
    }

```

```

String rawText = runOcr(imageStorageService.pathFor(image), request.language(), request.textLayout());
return toDraft(rawText);
}

private void requireOcrAccess(Authentication authentication) {
    boolean permitted = authentication.getAuthorities().stream()
        .anyMatch(authority -> "OCR_ENABLED".equals(authority.getAuthority())
            || "ROLE_ADMIN".equals(authority.getAuthority()));
    if (!permitted) {
        throw new OcrException("OCR is not available for this user.");
    }
}

private String validImageId(String imageId) {
    if (imageId == null) {
        throw new OcrException("Image was not found.");
    }

    try {
        return UUID.fromString(imageId).toString();
    } catch (IllegalArgumentException exception) {
        throw new OcrException("Image was not found.");
    }
}

private void validateParameters(OcrRequest request) {
    if (!LANGUAGES.contains(request.language()) || !TEXT_LAYOUTS.contains(request.textLayout())) {
        throw new OcrException("OCR parameters are invalid.");
    }
}

private String runOcr(Path imagePath, String language, int textLayout) {
    List<String> command = List.of(
        ocrCommand,
        imagePath.toString(),
        "stdout",
        "-l",
        language,
        "--psm",
        String.valueOf(textLayout)
    );

    try {
        Process process = new ProcessBuilder(command).redirectErrorStream(true).start();
        boolean completed = process.waitFor(30, TimeUnit.SECONDS);
    }
}

```

```

    if (!completed) {
        process.destroyForcibly();
        throw new OcrException("OCR timed out.");
    }

    String output = new String(process.getInputStream().readAllBytes(), StandardCharsets.UTF_8).trim();
    if (process.exitValue() != 0 || output.isEmpty()) {
        throw new OcrException("OCR could not extract text.");
    }
    return output;
} catch (IOException exception) {
    throw new OcrException("OCR could not be started.", exception);
} catch (InterruptedException exception) {
    Thread.currentThread().interrupt();
    throw new OcrException("OCR was interrupted.", exception);
}
}

private OcrDraft toDraft(String rawText) {
    String[] sections = rawText.split("\\R\\s*\\R", 2);
    String title = sections[0].trim();
    String description = sections.length == 1 ? title : sections[1].trim();
    return new OcrDraft(title, description, rawText);
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/preview/CommentPreviewController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/CommentPreviewController.java>

```
package com.securefromscratch.busybee.preview;

import java.util.Map;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
class CommentPreviewController {

    private final CommentPreviewGenerationService commentPreviewGenerationService;
    private final CommentPreviewDeletionService commentPreviewDeletionService;

    CommentPreviewController(
        CommentPreviewGenerationService commentPreviewGenerationService,
        CommentPreviewDeletionService commentPreviewDeletionService
    ){
        this.commentPreviewGenerationService = commentPreviewGenerationService;
        this.commentPreviewDeletionService = commentPreviewDeletionService;
    }

    @PostMapping("/link-preview/comment")
    TaskLinkPreviewResponse generate(Authentication authentication, @RequestBody CommentPreviewRequest request) {
        return commentPreviewGenerationService.generate(request, authentication.getName());
    }

    @PostMapping("/link-preview/comment/delete")
    PreviewDeleteResponse delete(@RequestBody CommentPreviewRequest request) {
        return commentPreviewDeletionService.delete(request);
    }

    @ExceptionHandler(InvalidPreviewRequestException.class)
    ResponseEntity<Map<String, String>> invalidPreviewRequest(InvalidPreviewRequestException exception) {
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/preview/CommentPreviewDeletionService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/CommentPreviewDeletionService.java>

```
package com.securefromscratch.busybee.preview;

import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
class CommentPreviewDeletionService {

    private final CommentPreviewRepository commentPreviewRepository;

    CommentPreviewDeletionService(CommentPreviewRepository commentPreviewRepository) {
        this.commentPreviewRepository = commentPreviewRepository;
    }

    @Transactional
    PreviewDeleteResponse delete(CommentPreviewRequest request) {
        String commentId = TaskPreviewUrlSupport.validCommentId(request.commentId());
        String commentText = commentPreviewRepository.findCommentText(commentId)
            .orElseThrow(() -> new InvalidPreviewRequestException("Comment was not found."));
        if (!TaskPreviewUrlSupport.textContainsUrl(commentText, request.url())) {
            throw new InvalidPreviewRequestException("URL is not present in this comment.");
        }

        commentPreviewRepository.deleteCommentPreview(commentId, request.url());
        return new PreviewDeleteResponse(true);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/preview/CommentPreviewGenerationService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/CommentPreviewGenerationService.java>

```

package com.securefromscratch.busybee.preview;

import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
class CommentPreviewGenerationService {

    private final CommentPreviewRepository commentPreviewRepository;
    private final TaskPreviewMetadataFetcher taskPreviewMetadataFetcher;

    CommentPreviewGenerationService(
        CommentPreviewRepository commentPreviewRepository,
        TaskPreviewMetadataFetcher taskPreviewMetadataFetcher
    ){
        this.commentPreviewRepository = commentPreviewRepository;
        this.taskPreviewMetadataFetcher = taskPreviewMetadataFetcher;
    }

    @Transactional
    TaskLinkPreviewResponse generate(CommentPreviewRequest request, String username) {
        String commentId = TaskPreviewUrlSupport.validCommentId(request.commentId());
        String commentText = commentPreviewRepository.findCommentText(commentId)
            .orElseThrow(() -> new InvalidPreviewRequestException("Comment was not found."));
        if (!TaskPreviewUrlSupport.textContainsUrl(commentText, request.url())) {
            throw new InvalidPreviewRequestException("URL is not present in this comment.");
        }

        GeneratedTaskPreview generated = taskPreviewMetadataFetcher.fetch(request.url());
        commentPreviewRepository.replaceCommentPreview(commentId, generated, username);
        return new TaskLinkPreviewResponse(
            generated.url(),
            generated.title(),
            generated.description(),
            generated.image()
        );
    }
}

```

**busybee/src/main/java/com/securefromscratch/busybee/preview/
CommentPreviewListingService.java**

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/CommentPreviewListingService.java>

```
package com.securefromscratch.busybee.preview;

import java.util.List;

import org.springframework.stereotype.Service;

@Service
public class CommentPreviewListingService {

    private final CommentPreviewRepository commentPreviewRepository;

    CommentPreviewListingService(CommentPreviewRepository commentPreviewRepository) {
        this.commentPreviewRepository = commentPreviewRepository;
    }

    public List<TaskLinkPreviewResponse> listCommentPreviews(String commentId) {
        return commentPreviewRepository.findCommentPreviews(commentId).stream()
            .map(preview -> new TaskLinkPreviewResponse(
                preview.url(),
                preview.title(),
                preview.description(),
                preview.image()
            ))
            .toList();
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/preview/ CommentPreviewRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/CommentPreviewRepository.java>

```

package com.securefromscratch.busybee.preview;

import java.util.List;
import java.util.Optional;
import java.util.UUID;

import org.jooq.DSLContext;
import org.jooq.impl.DSL;
import org.springframework.stereotype.Repository;

@Repository
class CommentPreviewRepository {

    private static final org.jooq.Table<?> COMMENTS = DSL.table("comments");
    private static final org.jooq.Field<String> COMMENT_ID = DSL.field("comment_id", String.class);
    private static final org.jooq.Field<String> COMMENT_TEXT = DSL.field("text", String.class);
    private static final org.jooq.Table<?> LINK_PREVIEWS = DSL.table("link_previews");
    private static final org.jooq.Field<String> PREVIEW_ID = DSL.field("preview_id", String.class);
    private static final org.jooq.Field<String> PREVIEW_COMMENT_ID = DSL.field("comment_id", String.class);
    private static final org.jooq.Field<String> URL = DSL.field("url", String.class);
    private static final org.jooq.Field<String> TITLE = DSL.field("title", String.class);
    private static final org.jooq.Field<String> DESCRIPTION = DSL.field("description", String.class);
    private static final org.jooq.Field<String> IMAGE_URL = DSL.field("image_url", String.class);
    private static final org.jooq.Field<String> GENERATED_BY = DSL.field("generated_by", String.class);

    private final DSLContext dsl;

    CommentPreviewRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    Optional<String> findCommentText(String commentId) {
        return dsl.select(COMMENT_TEXT)
            .from(COMMENTS)
            .where(COMMENT_ID.eq(commentId))
            .fetchOptional(COMMENT_TEXT);
    }

    void replaceCommentPreview(String commentId, GeneratedTaskPreview preview, String username) {
        deleteCommentPreview(commentId, preview.url());
        dsl.insertInto(LINK_PREVIEWS)
            .columns(PREVIEW_ID, PREVIEW_COMMENT_ID, URL, TITLE, DESCRIPTION, IMAGE_URL, GENERATED_BY)
            .values(
                UUID.randomUUID().toString(),

```

```

        commentId,
        preview.url(),
        preview.title(),
        preview.description(),
        preview.image(),
        username
    )
    .execute();
}

void deleteCommentPreview(String commentId, String url) {
    dsl.deleteFrom(LINK_PREVIEWS)
        .where(PREVIEW_COMMENT_ID.eq(commentId).and(URL.eq(url)))
        .execute();
}

List<StoredCommentPreview> findCommentPreviews(String commentId) {
    return dsl.select(URL, TITLE, DESCRIPTION, IMAGE_URL)
        .from(LINK_PREVIEWS)
        .where(PREVIEW_COMMENT_ID.eq(commentId))
        .orderBy(URL.asc())
        .fetch(record -> new StoredCommentPreview(
            record.get(URL),
            record.get(TITLE),
            record.get(DESCRIPTION),
            record.get(IMAGE_URL)
        ));
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/preview/CommentPreviewRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/CommentPreviewRequest.java>

```

package com.securefromscratch.busybee.preview;

record CommentPreviewRequest(String commentId, String url) {
}

```

busybee/src/main/java/com/securefromscratch/busybee/preview/GeneratedTaskPreview.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/GeneratedTaskPreview.java>

```
package com.securefromscratch.busybee.preview;

record GeneratedTaskPreview(String url, String title, String description, String image) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/preview/InvalidPreviewRequestException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/InvalidPreviewRequestException.java>

```
package com.securefromscratch.busybee.preview;

class InvalidPreviewRequestException extends RuntimeException {

    InvalidPreviewRequestException(String message) {
        super(message);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/preview/PreviewDeleteResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/PreviewDeleteResponse.java>

```
package com.securefromscratch.busybee.preview;

record PreviewDeleteResponse(boolean success) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/preview/StoredCommentPreview.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/StoredCommentPreview.java>

```
package com.securefromscratch.busybee.preview;

record StoredCommentPreview(String url, String title, String description, String image) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/preview/StoredTaskPreview.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/StoredTaskPreview.java>

```
package com.securefromscratch.busybee.preview;

record StoredTaskPreview(String url, String title, String description, String image) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/preview/TaskLinkPreviewResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/TaskLinkPreviewResponse.java>

```
package com.securefromscratch.busybee.preview;

public record TaskLinkPreviewResponse(String url, String title, String description, String image) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewDeleteRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewDeleteRequest.java>

```
package com.securefromscratch.busybee.preview;

record TaskPreviewDeleteRequest(String taskid, String url) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewDeletionController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewDeletionController.java>

```

package com.securefromscratch.busybee.preview;

import java.util.Map;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
class TaskPreviewDeletionController {

    private final TaskPreviewDeletionService taskPreviewDeletionService;

    TaskPreviewDeletionController(TaskPreviewDeletionService taskPreviewDeletionService) {
        this.taskPreviewDeletionService = taskPreviewDeletionService;
    }

    @PostMapping("/link-preview/task/delete")
    PreviewDeleteResponse deleteTaskPreview(@RequestBody TaskPreviewDeleteRequest request) {
        return taskPreviewDeletionService.deleteTaskPreview(request);
    }

    @ExceptionHandler(InvalidPreviewRequestException.class)
    ResponseEntity<Map<String, String>> invalidPreviewRequest(InvalidPreviewRequestException exception) {
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewDeletionService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewDeletionService.java>

```

package com.securefromscratch.busybee.preview;

import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
class TaskPreviewDeletionService {

    private final TaskPreviewRepository taskPreviewRepository;

    TaskPreviewDeletionService(TaskPreviewRepository taskPreviewRepository) {
        this.taskPreviewRepository = taskPreviewRepository;
    }

    @Transactional
    PreviewDeleteResponse deleteTaskPreview(TaskPreviewDeleteRequest request) {
        String taskId = TaskPreviewUrlSupport.validTaskId(request.taskid());
        String taskDescription = taskPreviewRepository.findTaskDescription(taskId)
            .orElseThrow(() -> new InvalidPreviewRequestException("Task was not found."));
        if (!TaskPreviewUrlSupport.taskContainsUrl(taskDescription, request.url())) {
            throw new InvalidPreviewRequestException("URL is not present in this task.");
        }

        taskPreviewRepository.deleteTaskPreview(taskId, request.url());
        return new PreviewDeleteResponse(true);
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewGenerateRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewGenerateRequest.java>

```

package com.securefromscratch.busybee.preview;

record TaskPreviewGenerateRequest(String taskid, String url) {
}

```

busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewGenerationController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewGenerationController.java>

```
package com.securefromscratch.busybee.preview;

import java.util.Map;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
class TaskPreviewGenerationController {

    private final TaskPreviewGenerationService taskPreviewGenerationService;

    TaskPreviewGenerationController(TaskPreviewGenerationService taskPreviewGenerationService) {
        this.taskPreviewGenerationService = taskPreviewGenerationService;
    }

    @PostMapping("/link-preview/task")
    TaskLinkPreviewResponse generate(
        Authentication authentication,
        @RequestBody TaskPreviewGenerateRequest request
    ){
        return taskPreviewGenerationService.generate(request, authentication.getName());
    }

    @ExceptionHandler(InvalidPreviewRequestException.class)
    ResponseEntity<Map<String, String>> invalidPreviewRequest(InvalidPreviewRequestException exception) {
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/preview/ TaskPreviewGenerationService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewGenerationService.java>

```

package com.securefromscratch.busybee.preview;

import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
class TaskPreviewGenerationService {

    private final TaskPreviewRepository taskPreviewRepository;
    private final TaskPreviewMetadataFetcher taskPreviewMetadataFetcher;

    TaskPreviewGenerationService(
        TaskPreviewRepository taskPreviewRepository,
        TaskPreviewMetadataFetcher taskPreviewMetadataFetcher
    ){
        this.taskPreviewRepository = taskPreviewRepository;
        this.taskPreviewMetadataFetcher = taskPreviewMetadataFetcher;
    }

    @Transactional
    TaskLinkPreviewResponse generate(TaskPreviewGenerateRequest request, String username) {
        String taskId = TaskPreviewUrlSupport.validTaskId(request.taskid());
        String taskDescription = taskPreviewRepository.findTaskDescription(taskId)
            .orElseThrow(() -> new InvalidPreviewRequestException("Task was not found."));
        if (!TaskPreviewUrlSupport.taskContainsUrl(taskDescription, request.url())) {
            throw new InvalidPreviewRequestException("URL is not present in this task.");
        }

        GeneratedTaskPreview generated = taskPreviewMetadataFetcher.fetch(request.url());
        taskPreviewRepository.replaceTaskPreview(taskId, generated, username);
        return new TaskLinkPreviewResponse(
            generated.url(),
            generated.title(),
            generated.description(),
            generated.image()
        );
    }
}

```

**busybee/src/main/java/com/securefromscratch/busybee/preview/
TaskPreviewListingService.java**

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewListingService.java>

```
package com.securefromscratch.busybee.preview;

import java.util.List;

import org.springframework.stereotype.Service;

@Service
public class TaskPreviewListingService {

    private final TaskPreviewRepository taskPreviewRepository;

    TaskPreviewListingService(TaskPreviewRepository taskPreviewRepository) {
        this.taskPreviewRepository = taskPreviewRepository;
    }

    public List<TaskLinkPreviewResponse> listTaskPreviews(String taskId) {
        return taskPreviewRepository.findTaskPreviews(taskId).stream()
            .map(preview -> new TaskLinkPreviewResponse(
                preview.url(),
                preview.title(),
                preview.description(),
                preview.image()
            ))
            .toList();
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/preview/ TaskPreviewMetadataFetcher.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewMetadataFetcher.java>

```

package com.securefromscratch.busybee.preview;

import java.net.InetAddress;
import java.net.URI;
import java.net.URISyntaxException;
import java.net.UnknownHostException;

import org.jsoup.Jsoup;
import org.jsoup.nodes.Document;
import org.jsoup.nodes.Element;
import org.springframework.stereotype.Service;

@Service
class TaskPreviewMetadataFetcher {

    GeneratedTaskPreview fetch(String url) {
        URI uri = validRemoteUri(url);
        rejectNonPublicHosts(uri.getHost());
        try {
            Document document = Jsoup.connect(uri.toString())
                .timeout(5_000)
                .maxBodySize(1_000_000)
                .followRedirects(false)
                .get();

            String title = nonBlank(document.title(), uri.toString());
            String description = metadata(document, "meta[name=description]", "meta[property=og:description]");
            String image = metadata(document, "meta[property=og:image]", "meta[name=twitter:image]");
            return new GeneratedTaskPreview(uri.toString(), title, description, image.isBlank() ? null : image);
        } catch (java.io.IOException exception) {
            throw new InvalidPreviewRequestException("Preview metadata could not be retrieved.");
        }
    }

    private URI validRemoteUri(String url) {
        try {
            URI uri = new URI(url);
            if (("http".equalsIgnoreCase(uri.getScheme()) || "https".equalsIgnoreCase(uri.getScheme()))
                && uri.getHost() != null) {
                return uri;
            }
        } catch (URISyntaxException exception) {
            throw new InvalidPreviewRequestException("URL is invalid.");
        }
    }
}

```

```

        throw new InvalidPreviewRequestException("URL is invalid.");
    }

    private void rejectNonPublicHosts(String host) {
        try {
            for (InetAddress address : InetAddress.getAllByName(host)) {
                if (address.isAnyLocalAddress()
                    || address.isLoopbackAddress()
                    || address.isLinkLocalAddress()
                    || address.isSiteLocalAddress()
                    || address.isMulticastAddress()) {
                    throw new InvalidPreviewRequestException("URL host is not allowed.");
                }
            }
        } catch (UnknownHostException exception) {
            throw new InvalidPreviewRequestException("URL host could not be resolved.");
        }
    }

    private String metadata(Document document, String primarySelector, String fallbackSelector) {
        Element element = document.selectFirst(primarySelector);
        if (element == null) {
            element = document.selectFirst(fallbackSelector);
        }
        return element == null ? "" : element.attr("content").trim();
    }

    private String nonBlank(String value, String fallback) {
        return value == null || value.isBlank() ? fallback : value.trim();
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewRepository.java>

```

package com.securefromscratch.busybee.preview;

import java.util.Optional;
import java.util.List;
import java.util.UUID;

import org.jooq.DSLContext;
import org.jooq.impl.DSL;
import org.springframework.stereotype.Repository;

@Repository
class TaskPreviewRepository {

    private static final org.jooq.Table<?> TASKS = DSL.table("tasks");
    private static final org.jooq.Field<String> TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Field<String> DESCRIPTION = DSL.field("description", String.class);
    private static final org.jooq.Table<?> LINK_PREVIEWS = DSL.table("link_previews");
    private static final org.jooq.Field<String> PREVIEW_TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Field<String> URL = DSL.field("url", String.class);
    private static final org.jooq.Field<String> PREVIEW_ID = DSL.field("preview_id", String.class);
    private static final org.jooq.Field<String> TITLE = DSL.field("title", String.class);
    private static final org.jooq.Field<String> PREVIEW_DESCRIPTION = DSL.field("description", String.class);
    private static final org.jooq.Field<String> IMAGE_URL = DSL.field("image_url", String.class);
    private static final org.jooq.Field<String> GENERATED_BY = DSL.field("generated_by", String.class);

    private final DSLContext dsl;

    TaskPreviewRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    Optional<String> findTaskDescription(String taskId) {
        return dsl.select(DESCRIPTION)
            .from(TASKS)
            .where(TASK_ID.eq(taskId))
            .fetchOptional(DESCRIPTION);
    }

    void deleteTaskPreview(String taskId, String url) {
        dsl.deleteFrom(LINK_PREVIEWS)
            .where(PREVIEW_TASK_ID.eq(taskId).and(URL.eq(url)))
            .execute();
    }
}

```

```

void replaceTaskPreview(String taskId, GeneratedTaskPreview preview, String username) {
    deleteTaskPreview(taskId, preview.url());
    dsl.insertInto(LINK_PREVIEWS)
        .columns(PREVIEW_ID, PREVIEW_TASK_ID, URL, TITLE, PREVIEW_DESCRIPTION, IMAGE_URL, GENERATED_BY)
        .values(
            UUID.randomUUID().toString(),
            taskId,
            preview.url(),
            preview.title(),
            preview.description(),
            preview.image(),
            username
        )
        .execute();
}

List<StoredTaskPreview> findTaskPreviews(String taskId) {
    return dsl.select(URL, TITLE, PREVIEW_DESCRIPTION, IMAGE_URL)
        .from(LINK_PREVIEWS)
        .where(PREVIEW_TASK_ID.eq(taskId))
        .orderBy(URL.asc())
        .fetch(record -> new StoredTaskPreview(
            record.get(URL),
            record.get(TITLE),
            record.get(PREVIEW_DESCRIPTION),
            record.get(IMAGE_URL)
        ));
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewUrlSupport.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/preview/TaskPreviewUrlSupport.java>

```
package com.securefromscratch.busybee.preview;

import java.util.Locale;
import java.util.UUID;
import java.util.regex.Matcher;
import java.util.regex.Pattern;

final class TaskPreviewUrlSupport {

    private static final Pattern URL_PATTERN = Pattern.compile(
        "(?i)(https?:/[^\s]+|(?:(www\\.)?[a-z0-9][a-z0-9-]*\\.[a-z]{2,})(?:/[^\s]*)?)"
    );

    private static final String TRAILING_URL_PUNCTUATION = ".,;:!?]]";

    private TaskPreviewUrlSupport() {
    }

    static String validTaskId(String taskId) {
        return validId(taskId, "Task");
    }

    static String validCommentId(String commentId) {
        return validId(commentId, "Comment");
    }

    private static String validId(String id, String resourceName) {
        if (id == null) {
            throw new InvalidPreviewRequestException(resourceName + " ID is invalid.");
        }

        try {
            return UUID.fromString(id).toString();
        } catch (IllegalArgumentException exception) {
            throw new InvalidPreviewRequestException(resourceName + " ID is invalid.");
        }
    }

    static boolean taskContainsUrl(String text, String requestedUrl) {
        return textContainsUrl(text, requestedUrl);
    }

    static boolean textContainsUrl(String text, String requestedUrl) {
        Matcher matcher = URL_PATTERN.matcher(text);
        while (matcher.find()) {
```

```

        if (normalizedUrl(matcher.group()).equals(normalizedUrl(requestedUrl))) {
            return true;
        }
    }
    return false;
}

static String normalizedUrl(String value) {
    if (value == null || value.isBlank()) {
        throw new InvalidPreviewRequestException("URL is invalid.");
    }

    String normalized = trimTrailingPunctuation(value.trim()).toLowerCase(Locale.ROOT);
    if (normalized.startsWith("https://")) {
        normalized = normalized.substring("https://".length());
    } else if (normalized.startsWith("http://")) {
        normalized = normalized.substring("http://".length());
    }
    if (normalized.startsWith("www.")) {
        normalized = normalized.substring("www.".length());
    }
    if (normalized.endsWith("/")) {
        normalized = normalized.substring(0, normalized.length() - 1);
    }
    if (normalized.isBlank() || !normalized.contains(".")) {
        throw new InvalidPreviewRequestException("URL is invalid.");
    }
    return normalized;
}

private static String trimTrailingPunctuation(String value) {
    int end = value.length();
    while (end > 0 && TRAILING_URL_PUNCTUATION.indexOf(value.charAt(end - 1)) >= 0) {
        end--;
    }
    return value.substring(0, end);
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/session/AddedSessionAccount.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/session/AddedSessionAccount.java>

```
package com.securefromscratch.busybee.session;
```

```
record AddedSessionAccount(String slot) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/session/ DemoSessionAuthenticationFilter.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/session/DemoSessionAuthenticationFilter.java>

```

package com.securefromscratch.busybee.session;

import java.io.IOException;

import jakarta.servlet.FilterChain;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.Cookie;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

import org.springframework.http.ResponseCookie;
import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.security.core.userdetails.UserDetails;
import org.springframework.security.core.userdetails.UserDetailsService;
import org.springframework.stereotype.Component;
import org.springframework.web.filter.OncePerRequestFilter;

@Component
public class DemoSessionAuthenticationFilter extends OncePerRequestFilter {

    private final DemoSessionProperties demoSessionProperties;
    private final UserDetailsService userDetailsService;

    DemoSessionAuthenticationFilter(
        DemoSessionProperties demoSessionProperties,
        UserDetailsService userDetailsService
    ) {
        this.demoSessionProperties = demoSessionProperties;
        this.userDetailsService = userDetailsService;
    }

    @Override
    protected void doFilterInternal(
        HttpServletRequest request,
        HttpServletResponse response,
        FilterChain filterChain
    ) throws ServletException, IOException {
        String sessionId = configuredSessionId(request);
        String username = sessionId == null ? null : demoSessionProperties.getUsers().get(sessionId);
        if (username != null && SecurityContextHolder.getContext().getAuthentication() == null) {
            UserDetails user = userDetailsService.loadUserByUsername(username);
            SecurityContextHolder.getContext().setAuthentication(
                UsernamePasswordAuthenticationToken.authenticated(user, null, user.getAuthorities())
            );
        }
        filterChain.doFilter(request, response);
    }
}

```

```

);
response.addHeader("Set-Cookie", ResponseCookie.from("JSESSIONID", sessionId)
    .path("/")
    .httpOnly(true)
    .maxAge(demoSessionProperties.getCookieMaxAge())
    .build()
    .toString());
}
filterChain.doFilter(request, response);
}

private String configuredSessionId(HttpServletRequest request) {
    if (!demoSessionProperties.isEnabled() || request.getCookies() == null) {
        return null;
    }
    for (Cookie cookie : request.getCookies()) {
        if ("JSESSIONID".equals(cookie.getName())
            && demoSessionProperties.getUsers().containsKey(cookie.getValue())) {
            return cookie.getValue();
        }
    }
    return null;
}
}

```

busybee/src/main/java/com/securefromscratch/busybee/session/DemoSessionProperties.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/session/DemoSessionProperties.java>

```

package com.securefromscratch.busybee.session;

import java.time.Duration;
import java.util.HashMap;
import java.util.Map;

import org.springframework.boot.context.properties.ConfigurationProperties;

@ConfigurationProperties("busybee.demo.sessions")
public class DemoSessionProperties {

    private boolean enabled;
    private Duration cookieMaxAge = Duration.ofHours(12);
    private Map<String, String> users = new HashMap<>();

    public boolean isEnabled() {
        return enabled;
    }

    public void setEnabled(boolean enabled) {
        this.enabled = enabled;
    }

    public Duration getCookieMaxAge() {
        return cookieMaxAge;
    }

    public void setCookieMaxAge(Duration cookieMaxAge) {
        this.cookieMaxAge = cookieMaxAge;
    }

    public Map<String, String> getUsers() {
        return users;
    }

    public void setUsers(Map<String, String> users) {
        this.users = users == null ? new HashMap<>() : new HashMap<>(users);
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/session/InvalidSessionSlotException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/session/InvalidSessionSlotException.java>

```
package com.securefromscratch.busybee.session;

class InvalidSessionSlotException extends RuntimeException {

    InvalidSessionSlotException(String message) {
        super(message);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/session/SessionAccount.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/session/SessionAccount.java>

```
package com.securefromscratch.busybee.session;

record SessionAccount(String slot, String username, boolean active) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/session/SessionAccountController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/session/SessionAccountController.java>

```

package com.securefromscratch.busybee.session;

import java.util.List;
import java.util.Map;

import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
class SessionAccountController {

    private final SessionAccountService sessionAccountService;

    SessionAccountController(SessionAccountService sessionAccountService) {
        this.sessionAccountService = sessionAccountService;
    }

    @GetMapping("/session/accounts")
    List<SessionAccount> accounts(Authentication authentication, HttpServletRequest request) {
        return sessionAccountService.accounts(authentication, request);
    }

    @PostMapping("/session/add-user")
    AddedSessionAccount addUser(Authentication authentication, HttpServletRequest request) {
        return sessionAccountService.parkCurrentAccount(authentication, request);
    }

    @PostMapping("/session/switch")
    ResponseEntity<Void> switchAccount(
        Authentication authentication,
        @RequestBody SwitchSessionRequest request,
        HttpServletRequest httpRequest,
        HttpServletResponse httpResponse
    ) {
        sessionAccountService.switchAccount(authentication, request, httpRequest, httpResponse);
    }
}

```

```
        return ResponseEntity.ok().build();
    }

    @ExceptionHandler(InvalidSessionSlotException.class)
    ResponseEntity<Map<String, String>> invalidSlot(InvalidSessionSlotException exception) {
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/session/SessionAccountService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/session/SessionAccountService.java>

```

package com.securefromscratch.busybee.session;

import java.util.ArrayList;
import java.util.LinkedHashMap;
import java.util.List;
import java.util.Map;

import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

import org.springframework.security.authentication.UsernamePasswordAuthenticationToken;
import org.springframework.security.core.Authentication;
import org.springframework.security.core.context.SecurityContext;
import org.springframework.security.core.context.SecurityContextHolder;
import org.springframework.security.core.userdetails.UserDetails;
import org.springframework.security.core.userdetails.UserDetailsService;
import org.springframework.stereotype.Service;
import org.springframework.security.web.context.HttpSessionSecurityContextRepository;

@Service
class SessionAccountService {

    private static final String PARKED_ACCOUNTS_ATTRIBUTE = "busybee.parked-accounts";

    private final UserDetailsService userDetailsService;
    private final HttpSessionSecurityContextRepository securityContextRepository =
        new HttpSessionSecurityContextRepository();

    SessionAccountService(UserDetailsService userDetailsService) {
        this.userDetailsService = userDetailsService;
    }

    List<SessionAccount> accounts(Authentication authentication, HttpServletRequest request) {
        List<SessionAccount> accounts = new ArrayList<>();
        accounts.add(new SessionAccount("current", authentication.getName(), true));
        parkedAccounts(request.getSession(false)).forEach(
            (slot, username) -> accounts.add(new SessionAccount(slot, username, false))
        );
        return accounts;
    }

    AddedSessionAccount parkCurrentAccount(Authentication authentication, HttpServletRequest request) {
        Map<String, String> parkedAccounts = parkedAccounts(request.getSession(true));
    }

```

```

String slot = nextSlot(parkedAccounts);
parkedAccounts.put(slot, authentication.getName());
return new AddedSessionAccount(slot);
}

void switchAccount(
    Authentication authentication,
    SwitchSessionRequest request,
    HttpServletRequest httpRequest,
    HttpServletResponse httpResponse
){
    if (request.slot() == null || request.slot().isBlank()) {
        throw new InvalidSessionSlotException("Session slot is required.");
    }

    Map<String, String> parkedAccounts = parkedAccounts(httpRequest.getSession(true));
    String requestedUsername = parkedAccounts.remove(request.slot());
    if (requestedUsername == null) {
        throw new InvalidSessionSlotException("Session slot was not found.");
    }
    parkedAccounts.put(nextSlot(parkedAccounts), authentication.getName());

    UserDetails user = userDetailsService.loadUserByUsername(requestedUsername);
    SecurityContext context = SecurityContextHolder.createEmptyContext();
    context.setAuthentication(UsernamePasswordAuthenticationToken.authenticated(user, null, user.getAuthorities()));
    SecurityContextHolder.setContext(context);
    securityContextRepository.saveContext(context, httpRequest, httpResponse);
}

@SuppressWarnings("unchecked")
private Map<String, String> parkedAccounts(HttpSession session) {
    if (session == null) {
        return Map.of();
    }
    Object existing = session.getAttribute(PARKED_ACCOUNTS_ATTRIBUTE);
    if (existing instanceof Map<?, ?>) {
        return (Map<String, String>) existing;
    }
    Map<String, String> parkedAccounts = new LinkedHashMap<>();
    session.setAttribute(PARKED_ACCOUNTS_ATTRIBUTE, parkedAccounts);
    return parkedAccounts;
}

private String nextSlot(Map<String, String> parkedAccounts) {

```

```
int slot = 1;
while (parkedAccounts.containsKey(Integer.toString(slot))) {
    slot++;
}
return Integer.toString(slot);
}
```

busybee/src/main/java/com/securefromscratch/busybee/session/SwitchSessionRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/session/SwitchSessionRequest.java>

```
package com.securefromscratch.busybee.session;

record SwitchSessionRequest(String slot) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/settings/InvalidSummaryThresholdException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/settings/InvalidSummaryThresholdException.java>

```
package com.securefromscratch.busybee.settings;

class InvalidSummaryThresholdException extends RuntimeException {

    InvalidSummaryThresholdException() {
        super("Summary threshold must be between 5 and 15.");
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsController.java>

```
package com.securefromscratch.busybee.settings;

import java.util.Map;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
class UserSettingsController {

    private final UserSettingsService userSettingsService;

    UserSettingsController(UserSettingsService userSettingsService) {
        this.userSettingsService = userSettingsService;
    }

    @GetMapping("/user/settings")
    UserSettingsResponse getSettings(Authentication authentication) {
        return userSettingsService.getSettings(authentication.getName());
    }

    @PostMapping("/user/settings")
    UserSettingsResponse replaceSettings(Authentication authentication, @RequestBody UserSettingsRequest request) {
        return userSettingsService.replaceSettings(authentication.getName(), request);
    }

    @ExceptionHandler(InvalidSummaryThresholdException.class)
    ResponseEntity<Map<String, String>> invalidSummaryThreshold(InvalidSummaryThresholdException exception) {
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/settings/ UserSettingsRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsRepository.java>

```

package com.securefromscratch.busybee.settings;

import java.util.Optional;

import org.jooq.DSLContext;
import org.jooq.impl.DSL;
import org.springframework.stereotype.Repository;

@Repository
class UserSettingsRepository {

    private static final org.jooq.Table<?> USER_SETTINGS = DSL.table("user_settings");
    private static final org.jooq.Field<String> USERNAME = DSL.field("username", String.class);
    private static final org.jooq.Field<Integer> SUMMARY_THRESHOLD_COMMENTS =
        DSL.field("summary_threshold_comments", Integer.class);

    private final DSLContext dsl;

    UserSettingsRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    Optional<Integer> findSummaryThreshold(String username) {
        return dsl.select(SUMMARY_THRESHOLD_COMMENTS)
            .from(USER_SETTINGS)
            .where(USERNAME.eq(username))
            .fetchOptional(SUMMARY_THRESHOLD_COMMENTS);
    }

    void saveSummaryThreshold(String username, int summaryThresholdComments) {
        dsl.insertInto(USER_SETTINGS)
            .columns(USERNAME, SUMMARY_THRESHOLD_COMMENTS)
            .values(username, summaryThresholdComments)
            .onDuplicateKeyUpdate()
            .set(SUMMARY_THRESHOLD_COMMENTS, summaryThresholdComments)
            .execute();
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/settings/ UserSettingsRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsRequest.java>

```
package com.securefromscratch.busybee.settings;

public record UserSettingsRequest(Integer summaryThresholdComments) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/settings/ UserSettingsResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsResponse.java>

```
package com.securefromscratch.busybee.settings;

import java.util.Map;

public record UserSettingsResponse(int summaryThresholdComments, Map<String, Object> aiCredential) {

    static UserSettingsResponse withoutAiCredential(int summaryThresholdComments) {
        return new UserSettingsResponse(summaryThresholdComments, Map.of());
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/settings/ UserSettingsService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/settings/UserSettingsService.java>

```

package com.securefromscratch.busybee.settings;

import org.springframework.stereotype.Service;

@Service
public class UserSettingsService {

    private static final int DEFAULT_SUMMARY_THRESHOLD = 5;
    private static final int MINIMUM_SUMMARY_THRESHOLD = 5;
    private static final int MAXIMUM_SUMMARY_THRESHOLD = 15;

    private final UserSettingsRepository userSettingsRepository;

    UserSettingsService(UserSettingsRepository userSettingsRepository) {
        this.userSettingsRepository = userSettingsRepository;
    }

    public UserSettingsResponse getSettings(String username) {
        int threshold = userSettingsRepository.findSummaryThreshold(username)
            .orElse(DEFAULT_SUMMARY_THRESHOLD);
        return UserSettingsResponse.withoutAiCredential(threshold);
    }

    UserSettingsResponse replaceSettings(String username, UserSettingsRequest request) {
        Integer threshold = request.summaryThresholdComments();
        if (threshold == null || threshold < MINIMUM_SUMMARY_THRESHOLD || threshold > MAXIMUM_SUMMARY_THRESHOLD) {
            throw new InvalidSummaryThresholdException();
        }

        userSettingsRepository.saveSummaryThreshold(username, threshold);
        return UserSettingsResponse.withoutAiCredential(threshold);
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/task/InvalidTaskException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/InvalidTaskException.java>

```
package com.securefromscratch.busybee.task;

class InvalidTaskException extends RuntimeException {

    InvalidTaskException(String message) {
        super(message);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/ListedTask.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/ListedTask.java>

```
package com.securefromscratch.busybee.task;

import java.time.LocalDate;
import java.time.LocalDateTime;
import java.time.LocalTime;

record ListedTask(
    String taskId,
    String name,
    String description,
    LocalDate dueDate,
    LocalTime dueTime,
    String createdBy,
    LocalDateTime createdAt,
    boolean done
) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskCompletionController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskCompletionController.java>

```

package com.securefromscratch.busybee.task;

import java.util.Map;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
class TaskCompletionController {

    private final TaskCompletionService taskCompletionService;

    TaskCompletionController(TaskCompletionService taskCompletionService) {
        this.taskCompletionService = taskCompletionService;
    }

    @PostMapping("/done")
    TaskCompletionResponse markDone(@RequestBody TaskCompletionRequest request) {
        return taskCompletionService.markDone(request);
    }

    @ExceptionHandler({InvalidTaskException.class, UnknownTaskException.class})
    ResponseEntity<Map<String, String>> taskCompletionError(RuntimeException exception) {
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskCompletionRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskCompletionRepository.java>

```

package com.securefromscratch.busybee.task;

import org.jooq.DSLContext;
import org.jooq.impl.DSL;
import org.springframework.stereotype.Repository;

@Repository
class TaskCompletionRepository {

    private static final org.jooq.Table<?> TASKS = DSL.table("tasks");
    private static final org.jooq.Field<String> TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Field<Boolean> DONE = DSL.field("done", Boolean.class);

    private final DSLContext dsl;

    TaskCompletionRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    boolean markDone(String taskId) {
        dsl.update(TASKS)
            .set(DONE, true)
            .where(TASK_ID.eq(taskId))
            .execute();
        return dsl.fetchExists(dsl.selectOne().from(TASKS).where(TASK_ID.eq(taskId)));
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskCompletionRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskCompletionRequest.java>

```

package com.securefromscratch.busybee.task;

record TaskCompletionRequest(String taskId) {
}

```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskCompletionResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskCompletionResponse.java>

```
package com.securefromscratch.busybee.task;

record TaskCompletionResponse(boolean success) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskCompletionService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskCompletionService.java>

```

package com.securefromscratch.busybee.task;

import java.util.UUID;

import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
class TaskCompletionService {

    private final TaskCompletionRepository taskCompletionRepository;

    TaskCompletionService(TaskCompletionRepository taskCompletionRepository) {
        this.taskCompletionRepository = taskCompletionRepository;
    }

    @Transactional
    TaskCompletionResponse markDone(TaskCompletionRequest request) {
        String taskId = validTaskId(request.taskid());
        if (!taskCompletionRepository.markDone(taskId)) {
            throw new UnknownTaskException();
        }

        return new TaskCompletionResponse(true);
    }

    private String validTaskId(String taskId) {
        if (taskId == null) {
            throw new InvalidTaskException("Task ID is invalid.");
        }

        try {
            return UUID.fromString(taskId).toString();
        } catch (IllegalArgumentException exception) {
            throw new InvalidTaskException("Task ID is invalid.");
        }
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskCreateRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskCreateRequest.java>

```
package com.securefromscratch.busybee.task;

import java.time.LocalDate;
import java.time.LocalTime;
import java.util.List;

record TaskCreateRequest(
    String name,
    String desc,
    LocalDate dueDate,
    LocalTime dueTime,
    List<String> responsibilityOf
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskCreateResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskCreateResponse.java>

```
package com.securefromscratch.busybee.task;

record TaskCreateResponse(String taskid) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskCreationController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskCreationController.java>

```

package com.securefromscratch.busybee.task;

import java.util.Map;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
class TaskCreationController {

    private final TaskCreationService taskCreationService;

    TaskCreationController(TaskCreationService taskCreationService) {
        this.taskCreationService = taskCreationService;
    }

    @PostMapping("/create")
    TaskCreateResponse createTask(Authentication authentication, @RequestBody TaskCreateRequest request) {
        return taskCreationService.createTask(authentication.getName(), request);
    }

    @ExceptionHandler(InvalidTaskException.class)
    ResponseEntity<Map<String, String>> invalidTask(InvalidTaskException exception) {
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskCreationService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskCreationService.java>

```

package com.securefromscratch.busybee.task;

import java.time.LocalDate;
import java.time.LocalTime;
import java.util.List;
import java.util.UUID;

import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
class TaskCreationService {

    private final TaskRepository taskRepository;

    TaskCreationService(TaskRepository taskRepository) {
        this.taskRepository = taskRepository;
    }

    @Transactional
    TaskCreateResponse createTask(String username, TaskCreateRequest request) {
        if (request.name() == null || request.name().isBlank()) {
            throw new InvalidTaskException("Task name is required.");
        }

        LocalDate dueDate = request.dueDate();
        LocalTime dueTime = dueDate == null ? null : request.dueTime();
        List<String> responsibilityOf = request.responsibilityOf() == null
            ? List.of()
            : request.responsibilityOf().stream().map(String::trim).filter(value -> !value.isEmpty()).toList();
        String taskId = UUID.randomUUID().toString();
        TaskToCreate task = new TaskToCreate(
            taskId,
            request.name().trim(),
            request.desc() == null ? "" : request.desc(),
            dueDate,
            dueTime,
            username,
            responsibilityOf
        );

        taskRepository.create(task);
        return new TaskCreateResponse(taskId);
    }
}

```

```
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskListResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskListResponse.java>

```
package com.securefromscratch.busybee.task;  
  
import java.time.LocalDate;  
import java.time.LocalDateTime;  
import java.util.List;  
  
import com.securefromscratch.busybee.comment.CommentListResponse;  
import com.securefromscratch.busybee.ai.CommentSummaryResponse;  
import com.securefromscratch.busybee.preview.TaskLinkPreviewResponse;  
  
record TaskListResponse(  
    String taskid,  
    String name,  
    String desc,  
    LocalDate dueDate,  
    String dueTime,  
    String createdBy,  
    List<String> responsibilityOf,  
    LocalDateTime creationDatetime,  
    boolean done,  
    CommentSummaryResponse commentSummary,  
    List<TaskLinkPreviewResponse> linkPreviews,  
    List<CommentListResponse> comments  
) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskListingController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1>

271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskListingController.java>

```
package com.securefromscratch.busybee.task;

import java.util.List;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
class TaskListingController {

    private final TaskListingService taskListingService;

    TaskListingController(TaskListingService taskListingService) {
        this.taskListingService = taskListingService;
    }

    @GetMapping("/tasks")
    List<TaskListResponse> listTasks() {
        return taskListingService.listTasks();
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskListingRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskListingRepository.java>

```

package com.securefromscratch.busybee.task;

import java.time.LocalDate;
import java.time.LocalDateTime;
import java.time.LocalTime;
import java.util.List;

import org.jooq.DSLContext;
import org.jooq.impl.DSL;
import org.springframework.stereotype.Repository;

@Repository
class TaskListingRepository {

    private static final org.jooq.Table<?> TASKS = DSL.table("tasks");
    private static final org.jooq.Field<String> TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Field<String> NAME = DSL.field("name", String.class);
    private static final org.jooq.Field<String> DESCRIPTION = DSL.field("description", String.class);
    private static final org.jooq.Field<LocalDate> DUE_DATE = DSL.field("due_date", LocalDate.class);
    private static final org.jooq.Field<LocalTime> DUE_TIME = DSL.field("due_time", LocalTime.class);
    private static final org.jooq.Field<String> CREATED_BY = DSL.field("created_by", String.class);
    private static final org.jooq.Field<LocalDateTime> CREATED_AT = DSL.field("created_at", LocalDateTime.class);
    private static final org.jooq.Field<Boolean> DONE = DSL.field("done", Boolean.class);
    private static final org.jooq.Table<?> TASK_RESPONSIBILITIES = DSL.table("task_responsibilities");
    private static final org.jooq.Field<String> RESPONSIBILITY_TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Field<String> RESPONSIBLE_NAME = DSL.field("responsible_name", String.class);
    private static final org.jooq.Field<Integer> POSITION_INDEX = DSL.field("position_index", Integer.class);

    private final DSLContext dsl;

    TaskListingRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    List<ListedTask> findAll() {
        return dsl.select(TASK_ID, NAME, DESCRIPTION, DUE_DATE, DUE_TIME, CREATED_BY, CREATED_AT, DONE)
            .from(TASKS)
            .orderBy(CREATED_AT.desc(), TASK_ID.desc())
            .fetch(record -> new ListedTask(
                record.get(TASK_ID),
                record.get(NAME),
                record.get(DESCRIPTION),
                record.get(DUE_DATE),
                record.get(DUE_TIME),

```

```
        record.get(CREATED_BY),
        record.get(CREATED_AT),
        Boolean.TRUE.equals(record.get(DONE))
    ));
}

List<String> findResponsibilities(String taskId) {
    return dsl.select(RESPONSIBLE_NAME)
        .from(TASK_RESPONSIBILITIES)
        .where(RESPONSIBILITY_TASK_ID.eq(taskId))
        .orderBy(POSITION_INDEX.asc())
        .fetch(RESPONSIBLE_NAME);
}
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskListingService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskListingService.java>

```

package com.securefromscratch.busybee.task;

import java.util.List;

import com.securefromscratch.busybee.comment.CommentListingService;
import com.securefromscratch.busybee.ai.CommentSummaryListingService;
import com.securefromscratch.busybee.preview.TaskPreviewListingService;

import org.springframework.stereotype.Service;

@Service
class TaskListingService {

    private final TaskListingRepository taskListingRepository;
    private final CommentListingService commentListingService;
    private final TaskPreviewListingService taskPreviewListingService;
    private final CommentSummaryListingService commentSummaryListingService;

    TaskListingService(
        TaskListingRepository taskListingRepository,
        CommentListingService commentListingService,
        TaskPreviewListingService taskPreviewListingService,
        CommentSummaryListingService commentSummaryListingService
    ){
        this.taskListingRepository = taskListingRepository;
        this.commentListingService = commentListingService;
        this.taskPreviewListingService = taskPreviewListingService;
        this.commentSummaryListingService = commentSummaryListingService;
    }

    List<TaskListResponse> listTasks() {
        return taskListingRepository.findAll().stream()
            .map(task -> new TaskListResponse(
                task.taskId(),
                task.name(),
                task.description(),
                task.dueDate(),
                task.dueTime() == null ? null : task.dueTime().toString(),
                task.createdBy(),
                taskListingRepository.findResponsibilities(task.taskId()),
                task.createdAt(),
                task.done(),
                commentSummaryListingService.summaryFor(task.taskId()),
                taskPreviewListingService.listTaskPreviews(task.taskId()),
            ))
    }
}

```

```
        commentListingService.listComments(task.taskId())
    ))
    .toList();
}
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskRepository.java>

```

package com.securefromscratch.busybee.task;

import java.time.LocalDate;
import java.time.LocalTime;
import java.util.List;

import org.jooq.DSLContext;
import org.jooq.impl.DSL;
import org.springframework.stereotype.Repository;

@Repository
class TaskRepository {

    private static final org.jooq.Table<?> TASKS = DSL.table("tasks");
    private static final org.jooq.Field<String> TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Field<String> NAME = DSL.field("name", String.class);
    private static final org.jooq.Field<String> DESCRIPTION = DSL.field("description", String.class);
    private static final org.jooq.Field<LocalDate> DUE_DATE = DSL.field("due_date", LocalDate.class);
    private static final org.jooq.Field<LocalTime> DUE_TIME = DSL.field("due_time", LocalTime.class);
    private static final org.jooq.Field<String> CREATED_BY = DSL.field("created_by", String.class);
    private static final org.jooq.Table<?> TASK_RESPONSIBILITIES = DSL.table("task_responsibilities");
    private static final org.jooq.Field<String> RESPONSIBILITY_TASK_ID = DSL.field("task_id", String.class);
    private static final org.jooq.Field<String> RESPONSIBLE_NAME = DSL.field("responsible_name", String.class);
    private static final org.jooq.Field<Integer> POSITION_INDEX = DSL.field("position_index", Integer.class);

    private final DSLContext dsl;

    TaskRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    void create(TaskToCreate task) {
        dsl.insertInto(TASKS)
            .columns(TASK_ID, NAME, DESCRIPTION, DUE_DATE, DUE_TIME, CREATED_BY)
            .values(task.taskId(), task.name(), task.description(), task.dueDate(), task.dueTime(), task.createdBy())
            .execute();

        for (int index = 0; index < task.responsibilityOf().size(); index++) {
            dsl.insertInto(TASK_RESPONSIBILITIES)
                .columns(RESPONSIBILITY_TASK_ID, RESPONSIBLE_NAME, POSITION_INDEX)
                .values(task.taskId(), task.responsibilityOf().get(index), index)
                .execute();
        }
    }
}

```

```
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/TaskToCreate.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/TaskToCreate.java>

```
package com.securefromscratch.busybee.task;  
  
import java.time.LocalDate;  
import java.time.LocalTime;  
import java.util.List;  
  
record TaskToCreate(  
    String taskId,  
    String name,  
    String description,  
    LocalDate dueDate,  
    LocalTime dueTime,  
    String createdBy,  
    List<String> responsibilityOf  
) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/task/UnknownTaskException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/task/UnknownTaskException.java>

```
package com.securefromscratch.busybee.task;

class UnknownTaskException extends RuntimeException {

    UnknownTaskException() {
        super("Task was not found.");
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/CurrentUserController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/CurrentUserController.java>

```
package com.securefromscratch.busybee.user;

import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
class CurrentUserController {

    private final CurrentUserService currentUserService;

    CurrentUserController(CurrentUserService currentUserService) {
        this.currentUserService = currentUserService;
    }

    @GetMapping("/me")
    CurrentUserResponse currentUser(Authentication authentication) {
        return currentUserService.currentUser(authentication);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/CurrentUserResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/CurrentUserResponse.java>

```
package com.securefromscratch.busybee.user;

import java.util.List;

public record CurrentUserResponse(
    String username,
    boolean admin,
    List<String> entitlements,
    List<String> effectiveEntitlements
){
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/CurrentUserService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/CurrentUserService.java>

```

package com.securefromscratch.busybee.user;

import java.util.List;
import java.util.Set;

import com.securefromscratch.busybee.admin.UserEntitlementService;

import org.springframework.security.core.Authentication;
import org.springframework.stereotype.Service;

@Service
class CurrentUserService {

    private static final Set<String> SUPPORTED_ENTITLEMENTS = Set.of(
        "IMPORT_ENABLED",
        "EXPORT_ENABLED",
        "PAID_LEVEL_1",
        "AI_ENABLED",
        "OCR_ENABLED"
    );

    private final UserEntitlementService userEntitlementService;

    CurrentUserService(UserEntitlementService userEntitlementService) {
        this.userEntitlementService = userEntitlementService;
    }

    CurrentUserResponse currentUser(Authentication authentication) {
        List<String> databaseEntitlements = userEntitlementService.entitlementsFor(authentication.getName());
        List<String> authorityEntitlements = authentication.getAuthorities().stream()
            .map(authority -> authority.getAuthority())
            .filter(SUPPORTED_ENTITLEMENTS::contains)
            .sorted()
            .toList();
        List<String> entitlements = databaseEntitlements.isEmpty() ? authorityEntitlements : databaseEntitlements;
        boolean admin = authentication.getAuthorities().stream()
            .anyMatch(authority -> "ROLE_ADMIN".equals(authority.getAuthority()));
        List<String> effectiveEntitlements = admin
            ? SUPPORTED_ENTITLEMENTS.stream().sorted().toList()
            : entitlements;

        return new CurrentUserResponse(authentication.getName(), admin, entitlements, effectiveEntitlements);
    }
}

```

```
}  
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/DatabaseUserDetailsService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/DatabaseUserDetailsService.java>

```
package com.securefromscratch.busybee.user;  
  
import org.springframework.security.core.userdetails.User;  
import org.springframework.security.core.userdetails.UserDetails;  
import org.springframework.security.core.userdetails.UserDetailsService;  
import org.springframework.security.core.userdetails.UsernameNotFoundException;  
import org.springframework.stereotype.Service;  
  
@Service  
class DatabaseUserDetailsService implements UserDetailsService {  
  
    private final UserRepository userRepository;  
  
    DatabaseUserDetailsService(UserRepository userRepository) {  
        this.userRepository = userRepository;  
    }  
  
    @Override  
    public UserDetails loadUserByUsername(String username) {  
        RegisteredUser user = userRepository.findByUsername(username)  
            .orElseThrow(() -> new UsernameNotFoundException("Unknown user."));  
  
        return User.withUsername(user.username())  
            .password(user.passwordHash())  
            .roles("USER")  
            .build();  
    }  
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/Dup

uplicateUsernameException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/DuplicateUsernameException.java>

```
package com.securefromscratch.busybee.user;

public class DuplicateUsernameException extends RuntimeException {

    public DuplicateUsernameException() {
        super("Username is already registered.");
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/IncorrectCurrentPasswordException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/IncorrectCurrentPasswordException.java>

```
package com.securefromscratch.busybee.user;

class IncorrectCurrentPasswordException extends RuntimeException {

    IncorrectCurrentPasswordException() {
        super("Current password is incorrect.");
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/InvalidNewPasswordException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/InvalidNewPasswordException.java>

```
package com.securefromscratch.busybee.user;

class InvalidNewPasswordException extends RuntimeException {

    InvalidNewPasswordException() {
        super("New password is required.");
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/InvalidRegistrationException.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/InvalidRegistrationException.java>

```
package com.securefromscratch.busybee.user;

class InvalidRegistrationException extends RuntimeException {

    InvalidRegistrationException() {
        super("Username and password are required.");
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/PasswordChangeController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/PasswordChangeController.java>

```

package com.securefromscratch.busybee.user;

import java.util.Map;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.PutMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
class PasswordChangeController {

    private final PasswordChangeService passwordChangeService;

    PasswordChangeController(PasswordChangeService passwordChangeService) {
        this.passwordChangeService = passwordChangeService;
    }

    @PutMapping("/user/password")
    PasswordChangeResponse changePassword(Authentication authentication, @RequestBody PasswordChangeRequest request) {
        return passwordChangeService.changePassword(authentication.getName(), request);
    }

    @ExceptionHandler({IncorrectCurrentPasswordException.class, InvalidNewPasswordException.class})
    ResponseEntity<Map<String, String>> passwordChangeError(RuntimeException exception) {
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/user/PasswordChangeRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/PasswordChangeRequest.java>

```
package com.securefromscratch.busybee.user;
```

```
record PasswordChangeRequest(String currentPassword, String newPassword) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/PasswordChangeResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/PasswordChangeResponse.java>

```
package com.securefromscratch.busybee.user;
```

```
record PasswordChangeResponse(boolean success) {  
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/PasswordChangeService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/PasswordChangeService.java>

```

package com.securefromscratch.busybee.user;

import org.springframework.security.crypto.password.PasswordEncoder;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;

@Service
class PasswordChangeService {

    private final UserRepository userRepository;
    private final PasswordEncoder passwordEncoder;

    PasswordChangeService(UserRepository userRepository, PasswordEncoder passwordEncoder) {
        this.userRepository = userRepository;
        this.passwordEncoder = passwordEncoder;
    }

    @Transactional
    PasswordChangeResponse changePassword(String username, PasswordChangeRequest request) {
        if (request.newPassword() == null || request.newPassword().isBlank()) {
            throw new InvalidNewPasswordException();
        }

        RegisteredUser user = userRepository.findByUsername(username)
            .orElseThrow(IncorrectCurrentPasswordException::new);

        if (request.currentPassword() == null || !passwordEncoder.matches(request.currentPassword(), user.passwordHash())) {
            throw new IncorrectCurrentPasswordException();
        }

        String replacementHash = passwordEncoder.encode(request.newPassword());
        if (userRepository.replacePasswordHash(username, user.passwordHash(), replacementHash) != 1) {
            throw new IncorrectCurrentPasswordException();
        }

        return new PasswordChangeResponse(true);
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/user/RegisteredUser.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/RegisteredUser.java>

```
package com.securefromscratch.busybee.user;

record RegisteredUser(String username, String passwordHash) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/RegistrationController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/RegistrationController.java>

```

package com.securefromscratch.busybee.user;

import java.util.Map;

import org.springframework.http.HttpStatus;
import org.springframework.http.ResponseEntity;
import org.springframework.security.web.csrf.CsrfToken;
import org.springframework.web.bind.annotation.ExceptionHandler;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PostMapping;
import org.springframework.web.bind.annotation.RequestBody;
import org.springframework.web.bind.annotation.RestController;

@RestController
class RegistrationController {

    private final RegistrationService registrationService;

    RegistrationController(RegistrationService registrationService) {
        this.registrationService = registrationService;
    }

    @PostMapping("/register")
    RegistrationResponse register(@RequestBody RegistrationRequest request) {
        return registrationService.register(request);
    }

    @GetMapping("/gencsrfToken")
    Map<String, String> csrfToken(CsrfToken csrfToken) {
        return Map.of("token", csrfToken.getToken(), "headerName", csrfToken.getHeaderName());
    }

    @ExceptionHandler({DuplicateUsernameException.class, InvalidRegistrationException.class})
    ResponseEntity<Map<String, String>> registrationError(RuntimeException exception) {
        return ResponseEntity.status(HttpStatus.BAD_REQUEST).body(Map.of("error", exception.getMessage()));
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/user/RegistrationRequest.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1

271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/RegistrationRequest.java>

```
package com.securefromscratch.busybee.user;

public record RegistrationRequest(String username, String password) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/RegistrationResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/RegistrationResponse.java>

```
package com.securefromscratch.busybee.user;

public record RegistrationResponse(String redirectTo) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/RegistrationService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/RegistrationService.java>

```

package com.securefromscratch.busybee.user;

import org.springframework.security.crypto.password.PasswordEncoder;
import org.springframework.stereotype.Service;

@Service
class RegistrationService {

    private final UserRepository userRepository;
    private final PasswordEncoder passwordEncoder;

    RegistrationService(UserRepository userRepository, PasswordEncoder passwordEncoder) {
        this.userRepository = userRepository;
        this.passwordEncoder = passwordEncoder;
    }

    RegistrationResponse register(RegistrationRequest request) {
        if (request.username() == null || request.username().isBlank()
            || request.password() == null || request.password().isBlank()) {
            throw new InvalidRegistrationException();
        }

        userRepository.create(request.username().trim(), passwordEncoder.encode(request.password()));
        return new RegistrationResponse("/main/main.html");
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/user/UserLookupController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/UserLookupController.java>

```
package com.securefromscratch.busybee.user;

import java.util.List;

import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RequestParam;
import org.springframework.web.bind.annotation.RestController;

@RestController
class UserLookupController {

    private final UserLookupService userLookupService;

    UserLookupController(UserLookupService userLookupService) {
        this.userLookupService = userLookupService;
    }

    @GetMapping("/users/lookup")
    List<UserLookupResponse> findUsers(@RequestParam String query) {
        return userLookupService.findUsers(query);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/UserLookupResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/UserLookupResponse.java>

```
package com.securefromscratch.busybee.user;

public record UserLookupResponse(String username) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/UserLookupService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1

271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/UserLookupService.java>

```
package com.securefromscratch.busybee.user;

import java.util.List;

import org.springframework.stereotype.Service;

@Service
class UserLookupService {

    private static final int MAXIMUM_RESULTS = 20;

    private final UserRepository userRepository;

    UserLookupService(UserRepository userRepository) {
        this.userRepository = userRepository;
    }

    List<UserLookupResponse> findUsers(String query) {
        String normalizedQuery = query.trim();
        if (normalizedQuery.isEmpty()) {
            return List.of();
        }

        return userRepository.findUsernamesContaining(normalizedQuery, MAXIMUM_RESULTS).stream()
            .map(UserLookupResponse::new)
            .toList();
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/UserProfileController.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/UserProfileController.java>

```
package com.securefromscratch.busybee.user;

import org.springframework.security.core.Authentication;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController
class UserProfileController {

    private final UserProfileService userProfileService;

    UserProfileController(UserProfileService userProfileService) {
        this.userProfileService = userProfileService;
    }

    @GetMapping("/user/profile")
    UserProfileResponse profile(Authentication authentication) {
        return userProfileService.profile(authentication);
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/UserProfileResponse.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/UserProfileResponse.java>

```
package com.securefromscratch.busybee.user;

import java.util.List;

public record UserProfileResponse(
    String username,
    boolean admin,
    List<String> entitlements,
    List<String> effectiveEntitlements,
    UserProfileSettings settings,
    AuthenticationEndpoints authentication
){
    public record AuthenticationEndpoints(String loginUrl, String logoutUrl, String changePasswordUrl) {
    }
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/UserProfileService.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/UserProfileService.java>

```

package com.securefromscratch.busybee.user;

import org.springframework.security.core.Authentication;
import org.springframework.stereotype.Service;

import com.securefromscratch.busybee.ai.AiCredentialStatusService;
import com.securefromscratch.busybee.settings.UserSettingsResponse;
import com.securefromscratch.busybee.settings.UserSettingsService;

@Service
class UserProfileService {

    private final CurrentUserService currentUserService;
    private final UserSettingsService userSettingsService;
    private final AiCredentialStatusService aiCredentialStatusService;

    UserProfileService(
        CurrentUserService currentUserService,
        UserSettingsService userSettingsService,
        AiCredentialStatusService aiCredentialStatusService
    ) {
        this.currentUserService = currentUserService;
        this.userSettingsService = userSettingsService;
        this.aiCredentialStatusService = aiCredentialStatusService;
    }

    UserProfileResponse profile(Authentication authentication) {
        CurrentUserResponse user = currentUserService.currentUser(authentication);
        UserSettingsResponse storedSettings = userSettingsService.getSettings(user.username());
        UserProfileSettings settings = new UserProfileSettings(
            storedSettings.summaryThresholdComments(),
            aiCredentialStatusService.statusFor(authentication)
        );
        return new UserProfileResponse(
            user.username(),
            user.admin(),
            user.entitlements(),
            user.effectiveEntitlements(),
            settings,
            new UserProfileResponse.AuthenticationEndpoints("/login", "/logout", "/user/password")
        );
    }
}

```

busybee/src/main/java/com/securefromscratch/busybee/user/UserProfileSettings.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/UserProfileSettings.java>

```
package com.securefromscratch.busybee.user;

import com.securefromscratch.busybee.ai.AiCredentialStatusResponse;

public record UserProfileSettings(int summaryThresholdComments, AiCredentialStatusResponse aiCredential) {
}
```

busybee/src/main/java/com/securefromscratch/busybee/user/UserRepository.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/java/com/securefromscratch/busybee/user/UserRepository.java>

```

package com.securefromscratch.busybee.user;

import java.util.List;
import java.util.Optional;

import org.jooq.DSLContext;
import org.jooq.impl.DSL;
import org.springframework.dao.DuplicateKeyException;
import org.springframework.stereotype.Repository;

@Repository
class UserRepository {

    private static final org.jooq.Table<?> USERS = DSL.table("users");
    private static final org.jooq.Field<String> USERNAME = DSL.field("username", String.class);
    private static final org.jooq.Field<String> PASSWORD_HASH = DSL.field("password_hash", String.class);

    private final DSLContext dsl;

    UserRepository(DSLContext dsl) {
        this.dsl = dsl;
    }

    void create(String username, String passwordHash) {
        try {
            dsl.insertInto(USERS)
                .columns(USERNAME, PASSWORD_HASH)
                .values(username, passwordHash)
                .execute();
        } catch (DuplicateKeyException exception) {
            throw new DuplicateUsernameException();
        }
    }

    Optional<RegisteredUser> findByUsername(String username) {
        return dsl.select(USERNAME, PASSWORD_HASH)
            .from(USERS)
            .where(USERNAME.eq(username))
            .fetchOptional(record -> new RegisteredUser(
                record.get(USERNAME),
                record.get(PASSWORD_HASH)
            ));
    }
}

```

```
List<String> findUsernamesContaining(String query, int maximumResults) {
    return dsl.select(USERNAME)
        .from(USERS)
        .where(USERNAME.containsIgnoreCase(query))
        .orderBy(USERNAME.asc())
        .limit(maximumResults)
        .fetch(USERNAME);
}

int replacePasswordHash(String username, String currentPasswordHash, String replacementPasswordHash) {
    return dsl.update(USERS)
        .set(PASSWORD_HASH, replacementPasswordHash)
        .where(USERNAME.eq(username).and(PASSWORD_HASH.eq(currentPasswordHash)))
        .execute();
}
}
```

busybee/src/main/resources/application.properties

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/resources/application.properties>

```
spring.datasource.url=jdbc:mysql://192.168.48.1:3307/bindingtest?createDatabaseIfNotExist=true&allowPublicKeyRetrieval=true
spring.datasource.username=root
spring.datasource.password=123456
spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver

spring.flyway.locations=classpath:db/migration/mysql
spring.flyway.validate-on-migrate=false
spring.jooq.sql-dialect=MYSQL
busybee.file-storage.directory=${java.io.tmpdir}/busybee-uploads
busybee.ocr.command=../tesseract_mock_java/build/install/tesseract_mock_java/bin/tesseract_mock_java
```

busybee/src/main/resources/db/migration/mysql/V12__ensure_user_entitlements.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/resources/db/migration/mysql/V12__ensure_user_entitlements.sql>

```
CREATE TABLE IF NOT EXISTS user_entitlements (  
  username VARCHAR(80) NOT NULL,  
  entitlement_name VARCHAR(40) NOT NULL,  
  PRIMARY KEY (username, entitlement_name),  
  CONSTRAINT fk_user_entitlements_user FOREIGN KEY (username) REFERENCES users (username) ON DELETE CASCADE  
);
```

busybee/src/main/resources/db/migration/mysql/V13__create_comment_summaries.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/resources/db/migration/mysql/V13__create_comment_summaries.sql>

```
CREATE TABLE IF NOT EXISTS task_comment_summaries (  
  task_id CHAR(36) NOT NULL,  
  summary TEXT NOT NULL,  
  summarized_comment_count INT NOT NULL,  
  summarized_latest_comment_at TIMESTAMP NULL,  
  generated_by VARCHAR(80) NOT NULL,  
  credential_source VARCHAR(20) NOT NULL,  
  generated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  PRIMARY KEY (task_id),  
  CONSTRAINT fk_task_comment_summaries_task FOREIGN KEY (task_id) REFERENCES tasks (task_id) ON DELETE CASCADE,  
  CONSTRAINT fk_task_comment_summaries_user FOREIGN KEY (generated_by) REFERENCES users (username)  
);
```

busybee/src/main/resources/db/migration/mysql/V1__create_users.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/resources/db/migration/mysql/V1__create_users.sql>

```
CREATE TABLE users (  
  username VARCHAR(80) NOT NULL,  
  password_hash VARCHAR(100) NOT NULL,  
  enabled BOOLEAN NOT NULL DEFAULT TRUE,  
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  PRIMARY KEY (username)  
);
```

busybee/src/main/resources/db/migration/mysql/V2__create_user_settings.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/resources/db/migration/mysql/V2__create_user_settings.sql>

```
CREATE TABLE IF NOT EXISTS user_settings (  
  username VARCHAR(80) NOT NULL,  
  summary_threshold_comments INT NOT NULL DEFAULT 5,  
  updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP,  
  PRIMARY KEY (username),  
  CONSTRAINT fk_user_settings_username FOREIGN KEY (username) REFERENCES users (username) ON DELETE CASCADE,  
  CONSTRAINT chk_user_settings_summary_threshold_comments  
    CHECK (summary_threshold_comments BETWEEN 5 AND 15)  
);
```



busybee/src/main/resources/db/migration/mysql/V3__create_tasks.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/resources/db/migration/mysql/V3__create_tasks.sql>

```

CREATE TABLE IF NOT EXISTS tasks (
  task_id CHAR(36) NOT NULL,
  name VARCHAR(200) NOT NULL,
  description TEXT NOT NULL,
  due_date DATE NULL,
  due_time TIME NULL,
  created_by VARCHAR(80) NOT NULL,
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
  done BOOLEAN NOT NULL DEFAULT FALSE,
  PRIMARY KEY (task_id),
  CONSTRAINT fk_tasks_created_by FOREIGN KEY (created_by) REFERENCES users (username)
);

CREATE TABLE IF NOT EXISTS task_responsibilities (
  task_id CHAR(36) NOT NULL,
  responsible_name VARCHAR(120) NOT NULL,
  position_index INT NOT NULL,
  PRIMARY KEY (task_id, position_index),
  CONSTRAINT fk_responsibilities_task FOREIGN KEY (task_id) REFERENCES tasks (task_id) ON DELETE CASCADE
);

```

busybee/src/main/resources/db/migration/mysql/V4__create_comments.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/resources/db/migration/mysql/V4__create_comments.sql>

```
CREATE TABLE IF NOT EXISTS comments (  
  comment_id CHAR(36) NOT NULL,  
  task_id CHAR(36) NOT NULL,  
  parent_comment_id CHAR(36) NULL,  
  text TEXT NOT NULL,  
  indent INT NOT NULL DEFAULT 0,  
  created_by VARCHAR(80) NOT NULL,  
  created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  PRIMARY KEY (comment_id),  
  CONSTRAINT fk_comments_task FOREIGN KEY (task_id) REFERENCES tasks (task_id) ON DELETE CASCADE,  
  CONSTRAINT fk_comments_parent FOREIGN KEY (parent_comment_id) REFERENCES comments (comment_id) ON DELETE CASCADE,  
  CONSTRAINT fk_comments_created_by FOREIGN KEY (created_by) REFERENCES users (username)  
);
```

busybee/src/main/resources/db/migration/mysql/V5__add_image_file_storage.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55/terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/resources/db/migration/mysql/V5__add_image_file_storage.sql>

```
CREATE TABLE IF NOT EXISTS stored_files (  
  file_id CHAR(36) NOT NULL,  
  original_name VARCHAR(255) NOT NULL,  
  content_type VARCHAR(120) NOT NULL,  
  file_kind VARCHAR(20) NOT NULL,  
  storage_name VARCHAR(80) NOT NULL,  
  uploaded_by VARCHAR(80) NOT NULL,  
  uploaded_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  PRIMARY KEY (file_id),  
  CONSTRAINT fk_stored_files_uploaded_by FOREIGN KEY (uploaded_by) REFERENCES users (username)  
);
```

```
ALTER TABLE comments ADD COLUMN image_file_id CHAR(36) NULL;
```

```
ALTER TABLE comments ADD CONSTRAINT fk_comments_image  
  FOREIGN KEY (image_file_id) REFERENCES stored_files (file_id);
```

busybee/src/main/resources/db/migration/mysql/V6__add_comment_attachments.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/resources/db/migration/mysql/V6__add_comment_attachments.sql>

```
ALTER TABLE comments ADD COLUMN attachment_file_id CHAR(36) NULL;
```

```
ALTER TABLE comments ADD CONSTRAINT fk_comments_attachment  
FOREIGN KEY (attachment_file_id) REFERENCES stored_files (file_id);
```

busybee/src/main/resources/db/migration/mysql/V7__create_link_previews.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/resources/db/migration/mysql/V7__create_link_previews.sql>

```
CREATE TABLE IF NOT EXISTS link_previews (  
  preview_id CHAR(36) NOT NULL,  
  task_id CHAR(36) NULL,  
  comment_id CHAR(36) NULL,  
  url VARCHAR(2048) NOT NULL,  
  title VARCHAR(500) NOT NULL,  
  description VARCHAR(1000) NOT NULL,  
  image_url VARCHAR(2048) NULL,  
  generated_by VARCHAR(80) NOT NULL,  
  generated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,  
  PRIMARY KEY (preview_id),  
  CONSTRAINT fk_link_previews_task FOREIGN KEY (task_id) REFERENCES tasks (task_id) ON DELETE CASCADE,  
  CONSTRAINT fk_link_previews_comment FOREIGN KEY (comment_id) REFERENCES comments (comment_id) ON DELETE CASCADE,  
  CONSTRAINT fk_link_previews_generated_by FOREIGN KEY (generated_by) REFERENCES users (username),  
  CONSTRAINT chk_link_previews_owner CHECK (  
    (task_id IS NOT NULL AND comment_id IS NULL) OR (task_id IS NULL AND comment_id IS NOT NULL)  
  )  
);
```

busybee/src/main/resources/db/migration/mysql/V8__create_user_entitlements.sql

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/main/resources/db/migration/mysql/V8__create_user_entitlements.sql>

```
CREATE TABLE IF NOT EXISTS user_entitlements (  
  username VARCHAR(80) NOT NULL,  
  entitlement_name VARCHAR(40) NOT NULL,  
  PRIMARY KEY (username, entitlement_name),  
  CONSTRAINT fk_user_entitlements_user FOREIGN KEY (username) REFERENCES users (username) ON DELETE CASCADE  
);
```

busybee/src/test/java/com/securefromscratch/busybee/functionality/AuthenticationFeatureTests.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/test/java/com/securefromscratch/busybee/functionality/AuthenticationFeatureTests.java>

```

package com.securefromscratch.busybee.functionality;

import java.util.Map;
import java.util.UUID;

import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.boot.test.autoconfigure.AutoConfigureMockMvc;
import org.springframework.http.MediaType;
import org.springframework.mock.web.MockHttpSession;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.web.servlet.MockMvc;
import org.springframework.test.web.servlet.MvcResult;

import com.fasterxml.jackson.databind.ObjectMapper;

import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestPostProcessors.csrf;
import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestBuilders.formLogin;
import static org.springframework.security.test.web.servlet.response.SecurityMockMvcResultMatchers.authenticated;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.post;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.jsonPath;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;

@SpringBootTest
@AutoConfigureMockMvc
@ActiveProfiles({"mysql", "test"})
class AuthenticationFeatureTests {

    @Autowired
    private MockMvc mvc;

    @Autowired
    private ObjectMapper objectMapper;

    @Test
    void registeredUserCanLoginAndReadCurrentUser() throws Exception {
        String username = "authentication-" + UUID.randomUUID();
        String password = "password-123";
        String registration = objectMapper.writeValueAsString(Map.of("username", username, "password", password));

        mvc.perform(post("/register").with(csrf()).contentType(MediaType.APPLICATION_JSON).content(registration))
            .andExpect(status().isOk());
    }
}

```

```
MvcResult login = mvc.perform(formLogin().user(username).password(password))
    .andExpect(authenticated().withUsername(username))
    .andReturn();
MockHttpSession session = (MockHttpSession) login.getRequest().getSession(false);

mvc.perform(get("/me").session(session))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.username").value(username))
    .andExpect(jsonPath("$.admin").value(false))
    .andExpect(jsonPath("$.entitlements").isEmpty())
    .andExpect(jsonPath("$.effectiveEntitlements").isEmpty());
}
}
```

busybee/src/test/java/com/securefromscratch/busybee/functionality/BusyBeeFunctionalityTests.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/test/java/com/securefromscratch/busybee/functionality/BusyBeeFunctionalityTests.java>

```
package com.securefromscratch.busybee.functionality;

import java.io.IOException;
import java.net.InetSocketAddress;
import java.net.Socket;
import java.nio.charset.StandardCharsets;
import java.util.List;
import java.util.Map;
import java.util.UUID;

import org.junit.jupiter.api.Assertions;
import org.junit.jupiter.api.BeforeAll;
import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.boot.web.mvc.test.autoconfigure.AutoConfigureMockMvc;
import org.springframework.http.MediaType;
import org.springframework.mock.web.MockMultipartFile;
import org.springframework.security.core.authority.SimpleGrantedAuthority;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.web.servlet.MockMvc;
import org.springframework.test.web.servlet.MvcResult;

import jakarta.servlet.http.Cookie;

import com.fasterxml.jackson.databind.JsonNode;
import com.fasterxml.jackson.databind.ObjectMapper;
import com.fasterxml.jackson.databind.node.ObjectNode;

import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestPostProcessors.csrf;
import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestPostProcessors.user;
import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestBuilders.formLogin;
import static org.springframework.security.test.web.servlet.response.SecurityMockMvcResultMatchers.authenticated;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.get;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.delete;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.multipart;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.post;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.put;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.content;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.jsonPath;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;

@SpringBootTest
@AutoConfigureMockMvc
```

```

@ActiveProfiles({"mysql", "test"})
class BusyBeeFunctionalityTests {

    private static final String FALLBAK_OPENAI_API_KEY = "do not hardcode!";

    private static final String MYSQL_DOWN_MESSAGE = ""
        *****

        MySQL is down. Start it with:
        C:\Program Files\MySQL\MySQL Server 8.0\bin>.\mysqld.exe
        *****

        """,
        ,

    @Autowired
    private MockMvc mvc;

    @Autowired
    private ObjectMapper objectMapper;

    @BeforeAll
    static void mysqlIsRunning() {
        Assertions.assertDoesNotThrow(BusyBeeFunctionalityTests::connectToMysql, MYSQL_DOWN_MESSAGE);
    }

    private static void connectToMysql() throws IOException {
        try (Socket socket = new Socket()) {
            socket.connect(new InetSocketAddress("192.168.48.1", 3307), 1000);
        }
    }

    @Test
    void contextLoads() {
    }

    @Test
    void registerRejectsDuplicateUser() throws Exception {
        String username = uniqueUsername();

        register(username)
            .andExpect(status().isOk())
            .andExpect(jsonPath("$.redirectTo").value("/main/main.html"));

        register(username)
            .andExpect(status().isBadRequest())
            .andExpect(jsonPath("$.error").value("Username is already registered."));
    }
}

```

```

}

@Test
void adminCanSearchUsersAndSetEntitlements() throws Exception {
    String username = createUser();

    mvc.perform(get("/admin/users")
        .queryParams("query", username)
        .with(user("admin").roles("ADMIN")))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$[0].username").value(username));

    mvc.perform(post("/admin/user-entitlements")
        .with(csrf())
        .with(user("admin").roles("ADMIN"))
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of(
            "username", username,
            "entitlements", List.of("IMPORT_ENABLED", "OCR_ENABLED")
        ))))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.username").value(username))
        .andExpect(jsonPath("$.entitlements[0]").value("IMPORT_ENABLED"))
        .andExpect(jsonPath("$.entitlements[1]").value("OCR_ENABLED"));

    mvc.perform(get("/me")
        .with(user(username).authorities(new SimpleGrantedAuthority("ROLE_USER"))))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.effectiveEntitlements[0]").value("IMPORT_ENABLED"))
        .andExpect(jsonPath("$.effectiveEntitlements[1]").value("OCR_ENABLED"));
}

@Test
void userCanCreateCommentAndCompleteTask() throws Exception {
    String username = createUser();
    String taskId = createTask(username);

    addComment(username, taskId, "hello from integration")
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.commentid").exists());

    mvc.perform(post("/done")
        .with(csrf())
        .with(user(username).roles("USER")))

```

```

        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of("taskId", taskId)))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.success").value(true));

    JsonNode task = findTask(username, taskId);
    Assertions.assertTrue(task.get("done").asBoolean());
}

@Test
void commentImageCanBeFetchedAndConvertedToOcrDraft() throws Exception {
    String username = createUser();
    String taskId = createTask(username);
    String commentId = commentWithImage(username, taskId);
    JsonNode comment = findComment(commentId);
    String imageId = comment.get("image").asText();

    Assertions.assertEquals("note.png", comment.get("imageFilename").asText());
    Assertions.assertNotEquals(imageId, comment.get("imageFilename").asText());
    Assertions.assertTrue(comment.get("attachment").isNull());
    Assertions.assertTrue(comment.get("attachmentFilename").isNull());

    mvc.perform(get("/image")
        .queryParam("img", imageId)
        .with(user(username).roles("USER")))
        .andExpect(status().isOk())
        .andExpect(content().contentTypeCompatibleWith(MediaType.IMAGE_PNG));

    mvc.perform(post("/ocr/image")
        .with(csrf())
        .with(user(username).authorities(
            new SimpleGrantedAuthority("ROLE_USER"),
            new SimpleGrantedAuthority("OCR_ENABLED")
        ))
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of(
            "image", imageId,
            "language", "eng",
            "textLayout", 3
        ))))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.title").isNotEmpty())
        .andExpect(jsonPath("$.description").isNotEmpty())
        .andExpect(jsonPath("$.rawText").isNotEmpty());
}

```

```
}
```

```
@Test
```

```
void svgCommentImageCanBeUploadedAndFetched() throws Exception {
```

```
    String username = createUser();
```

```
    String taskId = createTask(username);
```

```
    String commentId = commentWithSvgImage(username, taskId);
```

```
    JsonNode comment = findComment(commentId);
```

```
    String imageId = comment.get("image").asText();
```

```
    Assertions.assertEquals("diagram.svg", comment.get("imageFilename").asText());
```

```
    Assertions.assertNotEquals(imageId, comment.get("imageFilename").asText());
```

```
    mvc.perform(get("/image")
```

```
        .queryParam("img", imageId)
```

```
        .with(user(username).roles("USER")))
```

```
        .andExpect(status().isOk())
```

```
        .andExpect(content().contentTypeCompatibleWith("image/svg+xml"));
```

```
}
```

```
@Test
```

```
void commentAttachmentCanBeFetchedAndIncludesOriginalFilename() throws Exception {
```

```
    String username = createUser();
```

```
    String taskId = createTask(username);
```

```
    String commentId = commentWithAttachment(username, taskId);
```

```
    JsonNode comment = findComment(commentId);
```

```
    String attachmentId = comment.get("attachment").asText();
```

```
    Assertions.assertEquals("meeting-notes.txt", comment.get("attachmentFilename").asText());
```

```
    Assertions.assertNotEquals(attachmentId, comment.get("attachmentFilename").asText());
```

```
    Assertions.assertTrue(comment.get("image").isNull());
```

```
    Assertions.assertTrue(comment.get("imageFilename").isNull());
```

```
    mvc.perform(get("/attachment")
```

```
        .queryParam("file", attachmentId)
```

```
        .with(user(username).roles("USER")))
```

```
        .andExpect(status().isOk())
```

```
        .andExpect(content().contentTypeCompatibleWith(MediaType.TEXT_PLAIN));
```

```
}
```

```
@Test
```

```
void aiCredentialSelectionUsesPersonalKeyBeforeServerKey() throws Exception {
```

```
    String username = createUser();
```

```

mvc.perform(get("/ai/credential")
    .with(user(username).roles("USER")))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.provider").value("gemini"))
    .andExpect(jsonPath("$.model").value("gemini-2.5-flash-lite"))
    .andExpect(jsonPath("$.providerType").value("GEMINI_FLASH"))
    .andExpect(jsonPath("$.personalCredentialConfigured").value(false))
    .andExpect(jsonPath("$.serverCredentialAvailable").value(true))
    .andExpect(jsonPath("$.serverCredentialAllowed").value(false))
    .andExpect(jsonPath("$.selection").value("UNAVAILABLE")));

mvc.perform(put("/ai/credential")
    .with(csrf())
    .with(user(username).roles("USER"))
    .contentType(MediaType.APPLICATION_JSON)
    .content(json(Map.of(
        "provider", "GEMINI_FLASH",
        "apiKey", "AlzaPersonalGeminiCredential123456789"
    ))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.personalCredentialConfigured").value(true))
    .andExpect(jsonPath("$.personalCredentialSuffix").value("6789"))
    .andExpect(jsonPath("$.selection").value("PERSONAL_KEY")));

if (hasOpenAiApiKey()) {
    String openAiApiKey = openAiApiKey();
    mvc.perform(put("/ai/credential")
        .with(csrf())
        .with(user(username).roles("USER"))
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of(
            "provider", "GPT_5_NANO",
            "apiKey", openAiApiKey
        ))))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.provider").value("openai"))
        .andExpect(jsonPath("$.model").value("gpt-5-nano"))
        .andExpect(jsonPath("$.providerType").value("GPT_5_NANO"))
        .andExpect(jsonPath("$.personalCredentialConfigured").value(true))
        .andExpect(jsonPath("$.personalCredentialSuffix").value(openAiApiKey.substring(openAiApiKey.length() - 4)))
        .andExpect(jsonPath("$.selection").value("PERSONAL_KEY")));
}

mvc.perform(delete("/ai/credential")

```

```

        .with(csrf())
        .with(user(username).roles("USER")))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.personalCredentialConfigured").value(false))
        .andExpect(jsonPath("$.selection").value("UNAVAILABLE"));
    }

```

@Test

```

void aiCredentialSelectionAllowsServerKeyForEntitledUsers() throws Exception {
    String username = createUser();

```

```

    mvc.perform(get("/ai/credential")
        .with(user(username).authorities(
            new SimpleGrantedAuthority("ROLE_USER"),
            new SimpleGrantedAuthority("AI_ENABLED")
        )))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.serverCredentialAvailable").value(true))
        .andExpect(jsonPath("$.serverCredentialAllowed").value(true))
        .andExpect(jsonPath("$.selection").value("SERVER_KEY"));
}

```

@Test

```

void aiSummarizesLongCommentThreadAndMarksItStaleAfterNewComment() throws Exception {

```

```

    String username = createUser();
    String taskId = createTask(username);
    for (int index = 1; index <= 5; index++) {
        addComment(username, taskId, "summary comment")
            .andExpect(status().isOk());
    }

```

```

    mvc.perform(post("/ai/task/comment-summary")
        .with(csrf())
        .with(user(username).authorities(
            new SimpleGrantedAuthority("ROLE_USER"),
            new SimpleGrantedAuthority("AI_ENABLED")
        )))
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of("taskId", taskId)))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.summary").value("AI summary of the comment thread."))
        .andExpect(jsonPath("$.summarizedCommentCount").value(5))
        .andExpect(jsonPath("$.currentCommentCount").value(5))
        .andExpect(jsonPath("$.stale").value(false));
}

```

```

addComment(username, taskId, "newer comment")
    .andExpect(status().isOk());

JsonNode task = findTask(username, taskId);
Assertions.assertTrue(task.get("commentSummary").get("stale").asBoolean());
Assertions.assertEquals(6, task.get("commentSummary").get("currentCommentCount").asInt());
}

@Test
void aiRefreshRequiresPersonalCredentialAndTaskAssistanceReturnsStructuredSuggestions() throws Exception {
    String username = createUser();
    String taskId = createTask(username);
    for (int index = 1; index <= 5; index++) {
        addComment(username, taskId, "personal summary comment")
            .andExpect(status().isOk());
    }

    mvc.perform(put("/ai/credential")
        .with(csrf())
        .with(user(username).roles("USER"))
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of(
            "provider", "GEMINI_FLASH",
            "apiKey", "AlzaPersonalGeminiCredential123456789"
        ))))
        .andExpect(status().isOk());

    mvc.perform(post("/ai/task/comment-summary/refresh")
        .with(csrf())
        .with(user(username).roles("USER"))
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of("taskId", taskId))))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.summary").value("AI summary of the comment thread."))
        .andExpect(jsonPath("$.credentialSource").value("PERSONAL_KEY"));

    mvc.perform(post("/ai/task/improve")
        .with(csrf())
        .with(user(username).roles("USER"))
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of(
            "title", "rough title",

```

```

        "description", "rough description"
    ))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.title").value("Improved task title"))
    .andExpect(jsonPath("$.description").value("Improved task description."));

mvc.perform(post("/ai/task/subtasks")
    .with(csrf())
    .with(user(username).roles("USER"))
    .contentType(MediaType.APPLICATION_JSON)
    .content(json(Map.of(
        "title", "rough title",
        "description", "rough description"
    ))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.subtasks[0]").value("First subtask"));

mvc.perform(post("/ai/task/ocr-structure")
    .with(csrf())
    .with(user(username).roles("USER"))
    .contentType(MediaType.APPLICATION_JSON)
    .content(json(Map.of("rawText", "OCR raw text"))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.title").value("OCR task title"))
    .andExpect(jsonPath("$.description").value("Structured from OCR text."));
}

@Test
void userProfileIncludesSettingsAiCredentialAndSecurityEndpoints() throws Exception {
    String username = createUser();

    mvc.perform(get("/user/profile")
        .with(user(username).authorities(
            new SimpleGrantedAuthority("ROLE_USER"),
            new SimpleGrantedAuthority("AI_ENABLED")
        )))
        .andExpect(status().isOk())
        .andExpect(jsonPath("$.username").value(username))
        .andExpect(jsonPath("$.settings.summaryThresholdComments").value(5))
        .andExpect(jsonPath("$.settings.aiCredential.selection").value("SERVER_KEY"))
        .andExpect(jsonPath("$.authentication.loginUrl").value("/login"))
        .andExpect(jsonPath("$.authentication.logoutUrl").value("/logout"))
        .andExpect(jsonPath("$.authentication.changePasswordUrl").value("/user/password"));
}

```

@Test

void userCanReplaceSettingsAndLookupUsersForAssignment() throws Exception {

String firstUsername = createUser();

String secondUsername = createUser();

```
mvc.perform(put("/user/settings")
    .with(csrf())
    .with(user(firstUsername).roles("USER"))
    .contentType(MediaType.APPLICATION_JSON)
    .content(json(Map.of("summaryThresholdComments", 7))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.summaryThresholdComments").value(7));
```

```
mvc.perform(get("/user/settings")
    .with(user(firstUsername).roles("USER")))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.summaryThresholdComments").value(7));
```

```
mvc.perform(get("/users/lookup")
    .queryParam("query", secondUsername)
    .with(user(firstUsername).roles("USER")))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$[0].username").value(secondUsername));
```

}

@Test

void userCanChangePasswordAfterCurrentPasswordVerification() throws Exception {

String username = createUser();

```
mvc.perform(put("/user/password")
    .with(csrf())
    .with(user(username).roles("USER"))
    .contentType(MediaType.APPLICATION_JSON)
    .content(json(Map.of(
        "currentPassword", "123456",
        "newPassword", "new-password-123"
    ))))
    .andExpect(status().isOk())
    .andExpect(jsonPath("$.success").value(true));
```

```
mvc.perform(formLogin().user(username).password("new-password-123"))
    .andExpect(authenticated().withUsername(username));
```

}

@Test

void configuredDemoSessionCookieAuthenticatesAndRefreshesExpiration() throws Exception {
 registerIfNeeded("demo_session_user");

 mvc.perform(get("/me")
 .cookie(new Cookie("JSESSIONID", "DEMOSESSIONID1234567890"))
 .andExpect(status().isOk())
 .andExpect(jsonPath("\$.username").value("demo_session_user"))
 .andExpect(result -> Assertions.assertTrue(
 result.getResponse().getHeaders("Set-Cookie").stream()
 .anyMatch(header -> header.contains("JSESSIONID=DEMOSESSIONID1234567890")
 && header.contains("Max-Age="))
));
}

@Test

void linkPreviewAcceptsBareDomainPresentInTaskTextWithTrailingPunctuation() throws Exception {

String username = createUser();

String taskId = createTask(
 username,

 "Bare link preview",

 "Read securefromscratch.com."

);

 mvc.perform(post("/link-preview/task/delete")
 .with(csrf())
 .with(user(username).roles("USER"))
 .contentType(MediaType.APPLICATION_JSON)
 .content(json(Map.of(
 "taskId", taskId,
 "url", "securefromscratch.com"
))))
 .andExpect(status().isOk());
}

@Test

void entitledUsersCanExportAndImport() throws Exception {

String username = createUser();

 createTask(username);

 mvc.perform(get("/extra/export")
 .with(user(username).authorities(
 new SimpleGrantedAuthority("ROLE_USER"),

```

        new SimpleGrantedAuthority("EXPORT_ENABLED")
    )))
    .andExpect(status().isOk());

MockMultipartFile importFile = new MockMultipartFile(
    "file",
    "tasks.ser",
    MediaType.APPLICATION_OCTET_STREAM_VALUE,
    json(List.of(Map.of(
        "name", "Imported task",
        "desc", "Imported description",
        "dueDate", "2099-01-01",
        "dueTime", "09:00",
        "responsibilityOf", List.of(username)
    ))).getBytes(StandardCharsets.UTF_8)
);

mvc.perform(multipart("/extra/import")
    .file(importFile)
    .with(csrf())
    .with(user(username).authorities(
        new SimpleGrantedAuthority("ROLE_USER"),
        new SimpleGrantedAuthority("IMPORT_ENABLED")
    )))
    .andExpect(status().isOk());
}

private org.springframework.test.web.servlet.ResultActions register(String username) throws Exception {
    return mvc.perform(post("/register")
        .with(csrf())
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of(
            "username", username,
            "password", "123456"
        ))));
}

private String createUser() throws Exception {
    String username = uniqueUsername();
    register(username)
        .andExpect(status().isOk());
    return username;
}

```

```
private String createTask(String username) throws Exception {
    return createTask(
        username,
        "Integration task",
        "A task created by the server test."
    );
}
```

```
private String createTask(String username, String name, String description) throws Exception {
    MvcResult result = mvc.perform(post("/create")
        .with(csrf())
        .with(user(username).roles("USER"))
        .contentType(MediaType.APPLICATION_JSON)
        .content(json(Map.of(
            "name", name,
            "desc", description,
            "dueDate", "2099-01-01",
            "dueTime", "09:00",
            "responsibilityOf", List.of(username)
        ))))
        .andExpect(status().isOk())
        .andReturn();
    return jsonTree(result).get("taskid").asText();
}
```

```
private org.springframework.test.web.servlet.ResultActions addComment(String username, String taskId, String text) throws Exception {
    MockMultipartFile fields = jsonPart("commentFields", commentFields(text, taskId));
    return mvc.perform(multipart("/comment")
        .file(fields)
        .with(csrf())
        .with(user(username).roles("USER")));
}
```

```
private String commentWithImage(String username, String taskId) throws Exception {
    MockMultipartFile fields = jsonPart("commentFields", commentFields("image comment", taskId));
    MockMultipartFile file = new MockMultipartFile(
        "file",
        "note.png",
        MediaType.IMAGE_PNG_VALUE,
        new byte[]{1, 2, 3, 4, 5, 6}
    );
    MvcResult result = mvc.perform(multipart("/comment")
        .file(fields)
        .file(file)
    );
}
```

```

        .with(csrf())
        .with(user(username).roles("USER")))
        .andExpect(status().isOk())
        .andReturn();
    return jsonTree(result).get("commentid").asText();
}

```

```

private String commentWithSvgImage(String username, String taskId) throws Exception {
    MockMultipartFile fields = jsonPart("commentFields", commentFields("svg image comment", taskId));
    MockMultipartFile file = new MockMultipartFile(
        "file",
        "diagram.svg",
        "image/svg+xml",
        "<svg xmlns='http://www.w3.org/2000/svg' viewBox='0 0 10 10'><circle cx='5' cy='5' r='4'/></svg>".
    );
    MvcResult result = mvc.perform(multipart("/comment")
        .file(fields)
        .file(file)
        .with(csrf())
        .with(user(username).roles("USER")))
        .andExpect(status().isOk())
        .andReturn();
    return jsonTree(result).get("commentid").asText();
}

```

```

private String commentWithAttachment(String username, String taskId) throws Exception {
    MockMultipartFile fields = jsonPart("commentFields", commentFields("attachment comment", taskId));
    MockMultipartFile file = new MockMultipartFile(
        "file",
        "meeting-notes.txt",
        MediaType.TEXT_PLAIN_VALUE,
        "notes from the meeting".getBytes(StandardCharsets.UTF_8)
    );
    MvcResult result = mvc.perform(multipart("/comment")
        .file(fields)
        .file(file)
        .with(csrf())
        .with(user(username).roles("USER")))
        .andExpect(status().isOk())
        .andReturn();
    return jsonTree(result).get("commentid").asText();
}

```

```

private ObjectNode commentFields(String text, String taskId) {

```

```

    ObjectNode fields = objectMapper.createObjectNode();
    fields.put("text", text);
    fields.put("taskid", taskId);
    fields.putNull("commentid");
    return fields;
}

private String findCommentImage(String commentId) throws Exception {
    JsonNode comment = findComment(commentId);
    return comment.get("image").asText();
}

private JsonNode findComment(String commentId) throws Exception {
    JsonNode tasksNode = jsonTree(mvc.perform(get("/tasks")
        .with(user("reader").roles("USER")))
        .andExpect(status().isOk())
        .andReturn());
    for (JsonNode task : tasksNode) {
        for (JsonNode comment : task.get("comments")) {
            if (commentId.equals(comment.get("commentid").asText())) {
                return comment;
            }
        }
    }
    Assertions.fail("Created comment was not found.");
    return objectMapper.createObjectNode();
}

private JsonNode findTask(String username, String taskId) throws Exception {
    JsonNode tasksNode = jsonTree(mvc.perform(get("/tasks")
        .with(user(username).roles("USER")))
        .andExpect(status().isOk())
        .andReturn());
    for (JsonNode task : tasksNode) {
        if (taskId.equals(task.get("taskid").asText())) {
            return task;
        }
    }
    Assertions.fail("Created task was not found.");
    return objectMapper.createObjectNode();
}

private MockMultipartFile jsonPart(String name, Object value) throws Exception {
    return new MockMultipartFile(

```

```

        name,
        "blob",
        MediaType.APPLICATION_JSON_VALUE,
        json(value).getBytes(StandardCharsets.UTF_8)
    );
}

private String json(Object value) throws Exception {
    return objectMapper.writeValueAsString(value);
}

private JsonNode jsonTree(MvcResult result) throws Exception {
    return objectMapper.readTree(result.getResponse().getContentAsString());
}

private static String uniqueUsername() {
    return UUID.randomUUID().toString();
}

private static String openAiApiKey() {
    String apiKey = System.getenv("OPENAI_API_KEY");
    // ALLOW NULL LITERAL:
    // old api: getenv could return null if the environment variable is not set, and we want to allow that for local developmen
    if (apiKey == null || apiKey.isBlank()) {
        return FALLBACK_OPENAI_API_KEY;
    }

    return apiKey;
}

private static boolean hasOpenAiApiKey() {
    return !FALLBACK_OPENAI_API_KEY.equals(openAiApiKey());
}

private void registerIfNeeded(String username) throws Exception {
    register(username)
        .andExpect(result -> Assertions.assertTrue(
            result.getResponse().getStatus() == 200
            || result.getResponse().getStatus() == 400
        ));
}
}

```

busybee/src/test/java/com/securefromscratch/busybee/functionality/RegistrationFeatureTests.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/test/java/com/securefromscratch/busybee/functionality/RegistrationFeatureTests.java>

```
package com.securefromscratch.busybee.functionality;

import java.util.Map;
import java.util.UUID;

import org.junit.jupiter.api.Test;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.boot.test.context.SpringBootTest;
import org.springframework.boot.web.mvc.test.autoconfigure.AutoConfigureMockMvc;
import org.springframework.http.MediaType;
import org.springframework.test.context.ActiveProfiles;
import org.springframework.test.web.servlet.MockMvc;

import com.fasterxml.jackson.databind.ObjectMapper;

import static org.springframework.security.test.web.servlet.request.SecurityMockMvcRequestPostProcessors.csrf;
import static org.springframework.test.web.servlet.request.MockMvcRequestBuilders.post;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.jsonPath;
import static org.springframework.test.web.servlet.result.MockMvcResultMatchers.status;

@SpringBootTest
@AutoConfigureMockMvc
@ActiveProfiles({"mysql", "test"})
class RegistrationFeatureTests {

    @Autowired
    private MockMvc mvc;

    @Autowired
    private ObjectMapper objectMapper;

    @Test
    void userCanRegisterAndDuplicateRegistrationIsReported() throws Exception {
        String username = "registration-" + UUID.randomUUID();
        String body = objectMapper.writeValueAsString(Map.of("username", username, "password", "123456"));

        mvc.perform(post("/register").with(csrf()).contentType(MediaType.APPLICATION_JSON).content(body))
            .andExpect(status().isOk())
            .andExpect(jsonPath("$.redirectTo").value("/main/main.html"));

        mvc.perform(post("/register").with(csrf()).contentType(MediaType.APPLICATION_JSON).content(body))
            .andExpect(status().isBadRequest())
            .andExpect(jsonPath("$.error").value("Username is already registered."));
    }
}
```

```
}  
}
```

busybee/src/test/resources/application-test.properties

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/busybee/src/test/resources/application-test.properties>

```
spring.datasource.url=jdbc:mysql://192.168.48.1:3307/bindingtest_test?createDatabaseIfNotExist=true&allowPublicKeyRetrieval=true
spring.datasource.username=root
spring.datasource.password=123456
spring.datasource.driver-class-name=com.mysql.cj.jdbc.Driver

spring.flyway.enabled=true
spring.flyway.locations=classpath:db/migration/mysql
spring.flyway.baseline-on-migrate=true
spring.flyway.validate-on-migrate=false

spring.jooq.sql-dialect=MYSQL
busybee.ai.gemini.api-key=AlzaTestServerCredentialForGeminiFlashLite
busybee.demo.sessions.enabled=true
busybee.demo.sessions.cookie-max-age=12h
busybee.demo.sessions.users[DEMOSESSIONID1234567890]=demo_session_user
```

gradle/wrapper/gradle-wrapper.properties

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/gradle/wrapper/gradle-wrapper.properties>

```
distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https://services.gradle.org/distributions/gradle-9.4.1-bin.zip
networkTimeout=10000
validateDistributionUrl=true
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
```

tesseract_mock_java/build.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/tesseract_mock_java/build.gradle.kts>

```
plugins {  
    application  
}  
  
group = "org.owasp.untrust"  
version = "0.1.0"  
  
java {  
    toolchain {  
        languageVersion = JavaLanguageVersion.of(21)  
    }  
}  
  
application {  
    mainClass = "org.owasp.untrust.tesseractmock.TesseractMock"  
}
```

tesseract_mock_java/settings.gradle.kts

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/tesseract_mock_java/settings.gradle.kts>

```
rootProject.name = "tesseract_mock_java"
```

tesseract_mock_java/src/main/java/org/owasp/untrust/tesseractmock/TesseractMock.java

Source:

<https://raw.githubusercontent.com/SecureFromScratch/BusyBee_generation_test_55terra_plain/1271837b7d6205536569c1f1f036ab67cd76a9e0/tesseract_mock_java/src/main/java/org/owasp/untrust/tesseractmock/TesseractMock.java>

```

package org.owasp.untrust.tesseractmock;

import java.nio.charset.StandardCharsets;
import java.nio.file.Files;
import java.nio.file.Path;
import java.util.List;

public final class TesseractMock {
    private static final List<String> OUTPUTS = List.of(
        "Buy printer paper",

        "Order two boxes of A4 printer paper for the office supply cabinet.",
        "Renew passport",

        "Fill the renewal form, attach a photo, and schedule the appointment.",
        "Pay utility bill",

        "Electricity bill is due this Friday. Confirm payment receipt afterward.",
        "Call project vendor",

        "Ask for the revised delivery estimate and request written confirmation.",
        "Prepare release notes",

        "Summarize fixed bugs, migration notes, and known limitations for review."
    );

    private TesseractMock() {}

    public static void main(String[] args) throws Exception {
        Arguments parsed = Arguments.parse(args);
        byte[] imageBytes = Files.readAllBytes(parsed.input());
        int outputIndex = Math.floorMod(imageBytes.length + parsed.language().hashCode() + parsed.pageSegmentationModel().hashCode(), OUTPUTS.size());
        System.out.print(OUTPUTS.get(outputIndex));
    }
}

```

```

}

private record Arguments(Path input, String outputBase, String language, int pageSegmentationMode) {
    private static Arguments parse(String[] args) {
        if (args.length != 6) {
            usage();
        }
        if (!"stdout".equals(args[1])) {
            usage();
        }
        if (!"-l".equals(args[2])) {
            usage();
        }
        if (args[3].isBlank()) {
            usage();
        }
        if (!"--psm".equals(args[4])) {
            usage();
        }
        return new Arguments(Path.of(args[0]), args[1], args[3], Integer.parseInt(args[5]));
    }

    private static void usage() {
        System.err.println("usage: tesseract <input-file> stdout -l <language> --psm <mode>");
        System.exit(2);
    }
}

```